



EPS 576 – Fall 2025

Infiltration Guest Lecture

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Outline

- Why we care about infiltration
- Water movement through soil – saturated conditions
- Water movement through soil – unsaturated conditions
- The shape of a typical infiltration curve
- What impacts infiltration? Test your intuition!
- Modeling infiltration – which model to choose?

Why do we care?

Roswell, NM (Oct. 2024)



Image source: Roswell Daily Record
https://www.rdrnews.com/news/local/after-the-flood-scenes-of-damage-across-the-region/article_ae75de74-8f26-11ef-bd35-53d9dcbef951.html

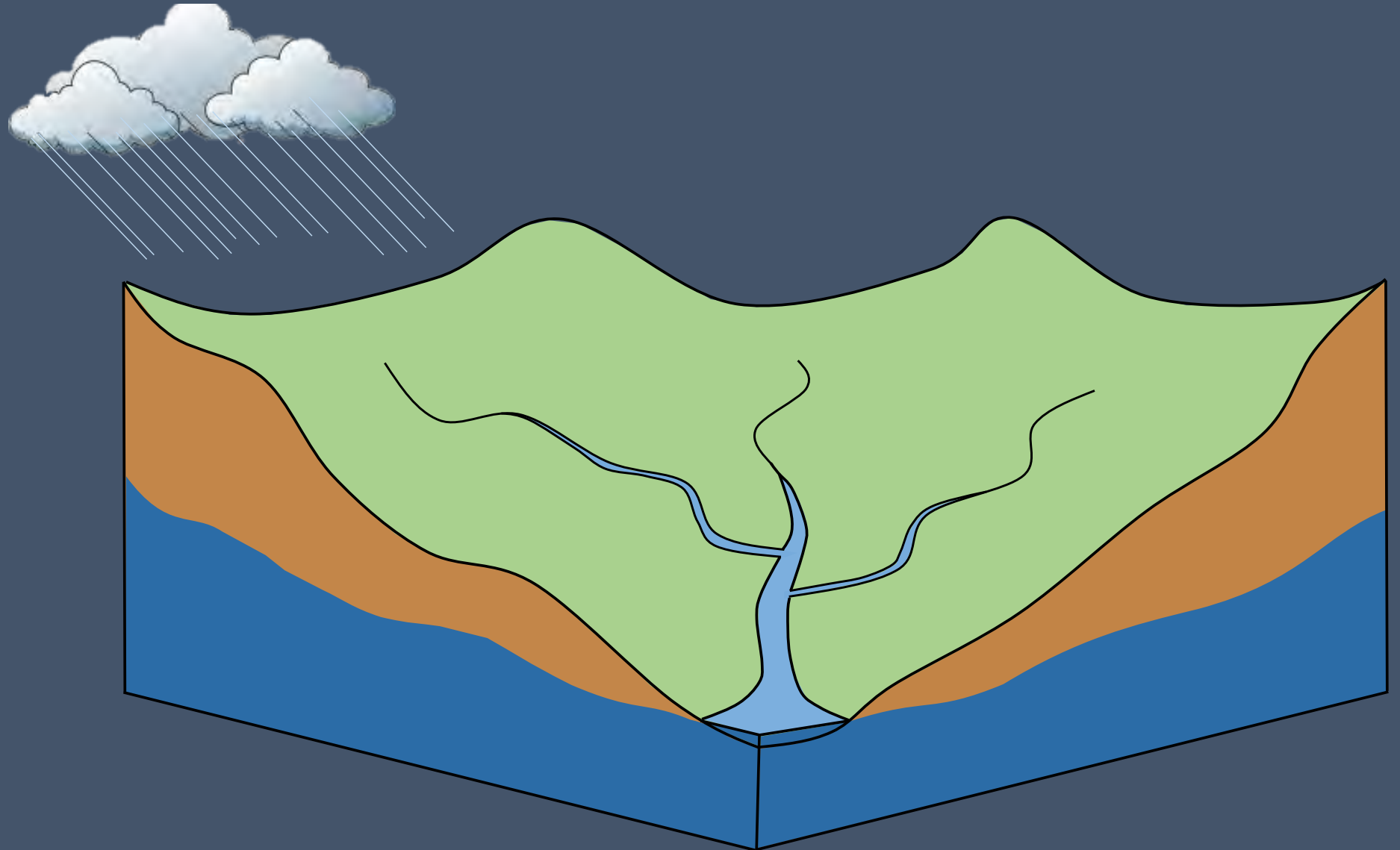
Valencia, Spain (Oct. 2024)



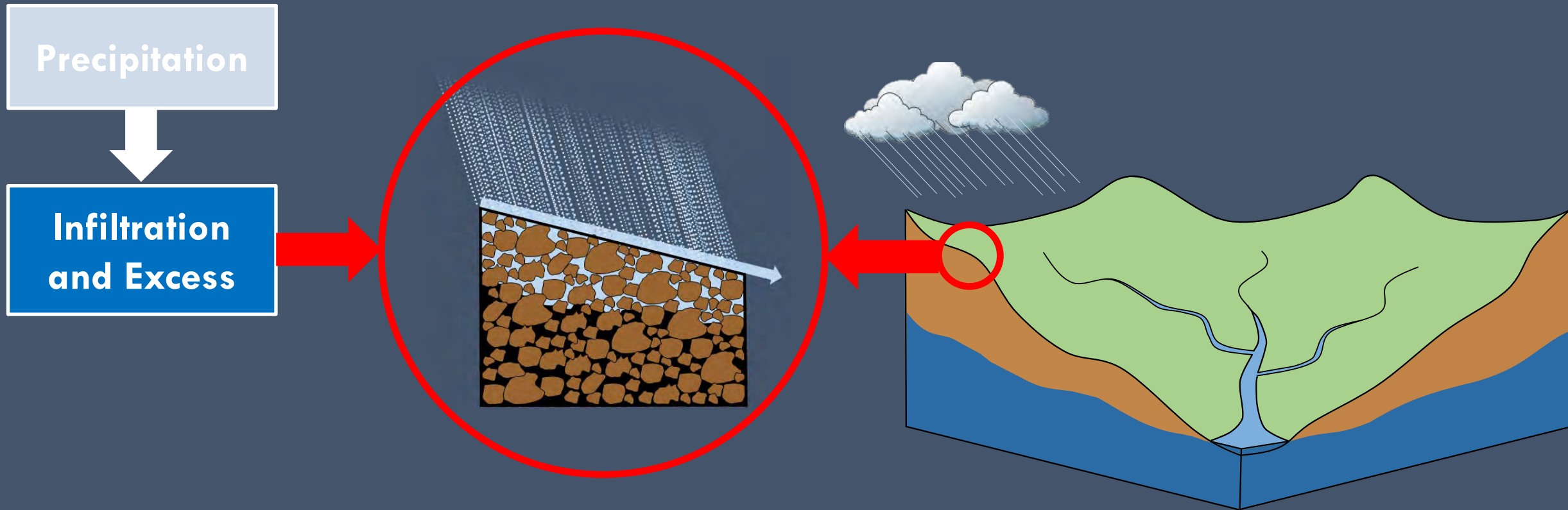
Image source: NPR
<https://www.npr.org/sections/the-picture-show/2024/10/30/g-s1-30937/valencia-spain-flooding-photos>

Conceptual watershed model

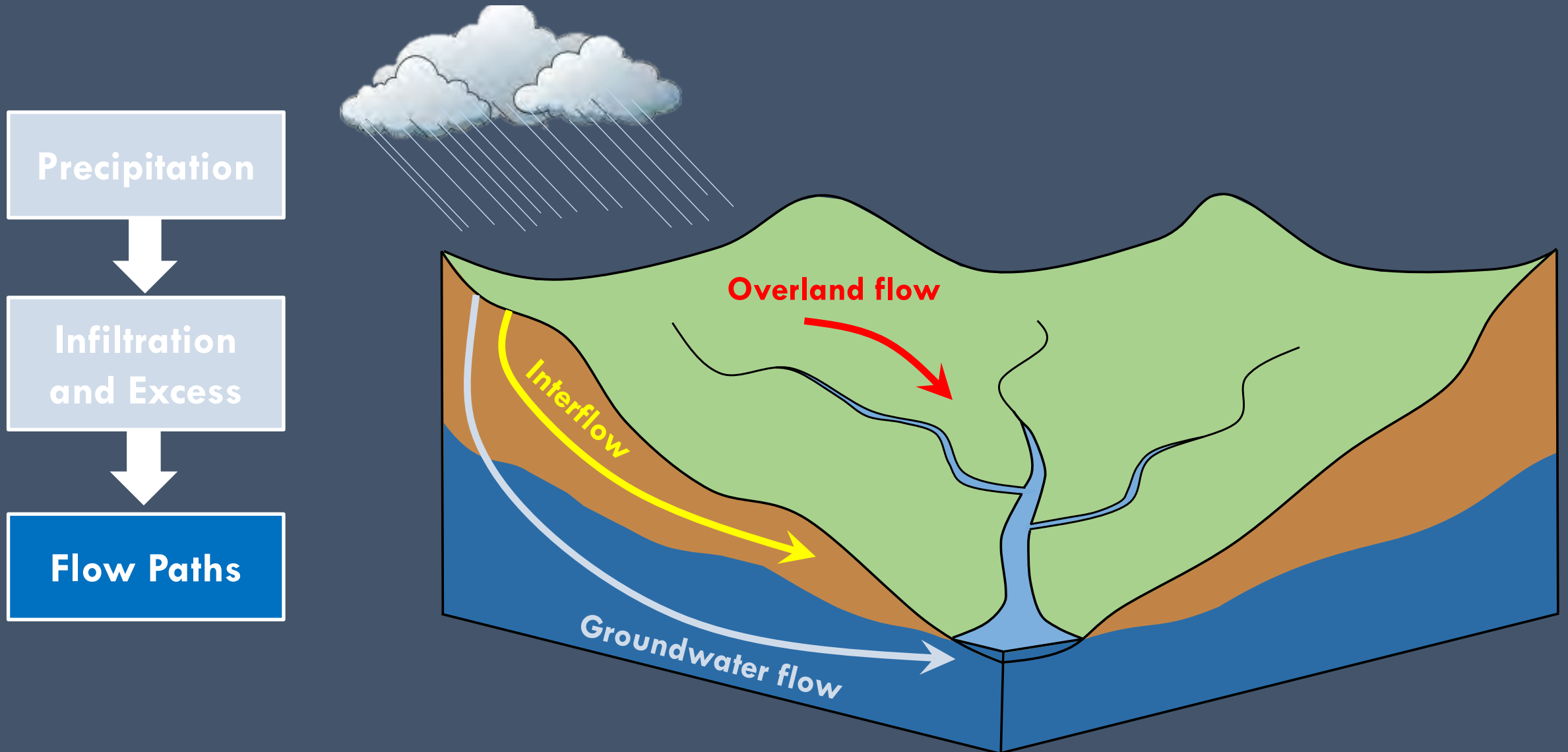
Precipitation



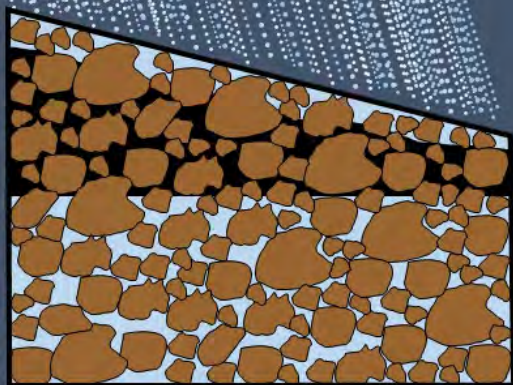
Conceptual watershed model



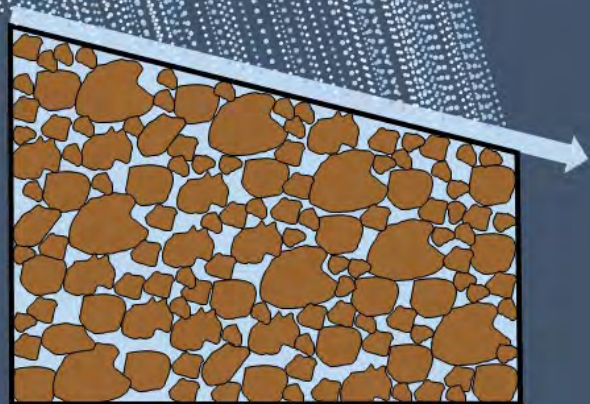
Conceptual model



saturated

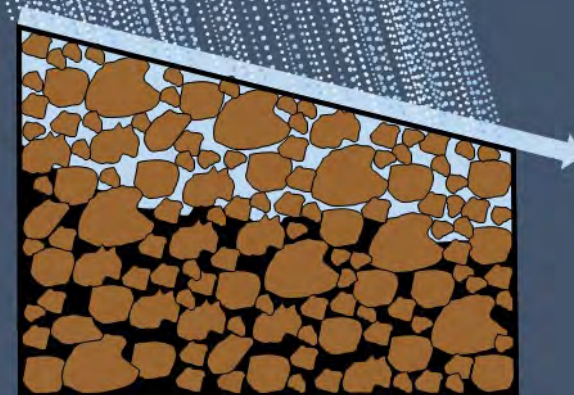
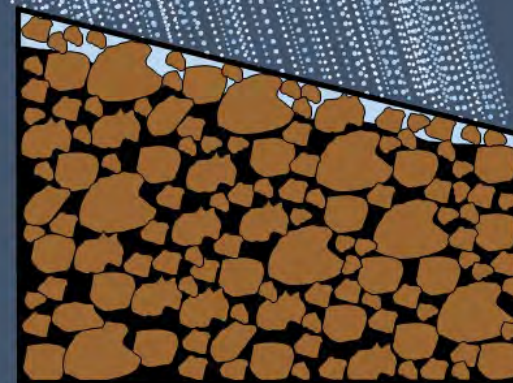


saturated



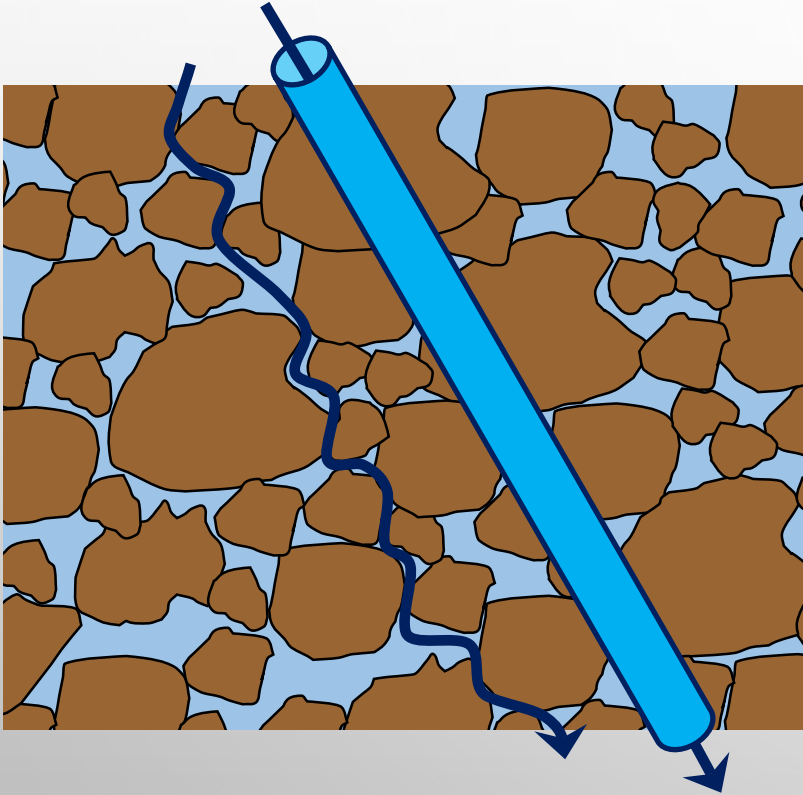
Saturation-excess overland flow

Rainfall rate
>
Infiltration rate



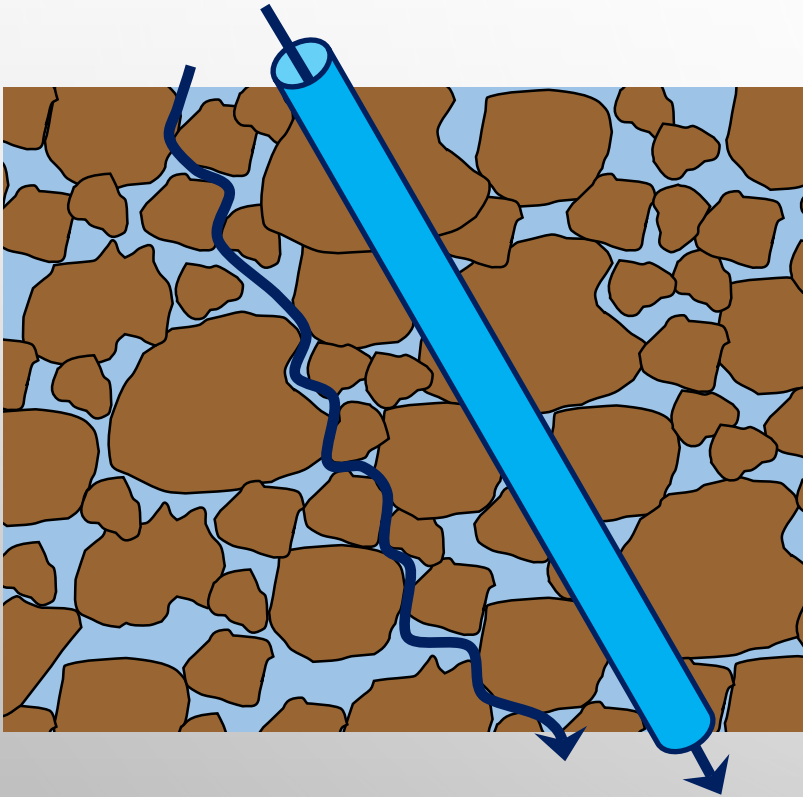
Infiltration-excess overland flow

Water movement through soil – saturated conditions



- Water movement through porous medium (e.g. sand) follows a convoluted path
- Very complex, so we use a conceptual model: flow through a bundle of capillary tubes

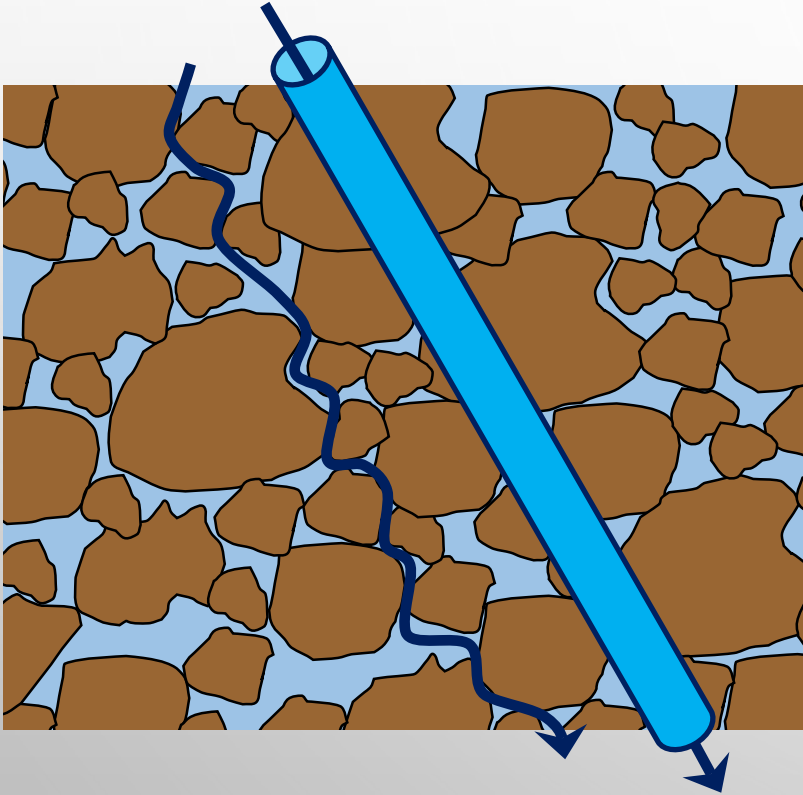
Water movement through soil – saturated conditions



- Darcy's law for flow through a porous medium:

$$q = -K \frac{dh}{dl}$$

Water movement through soil – saturated conditions

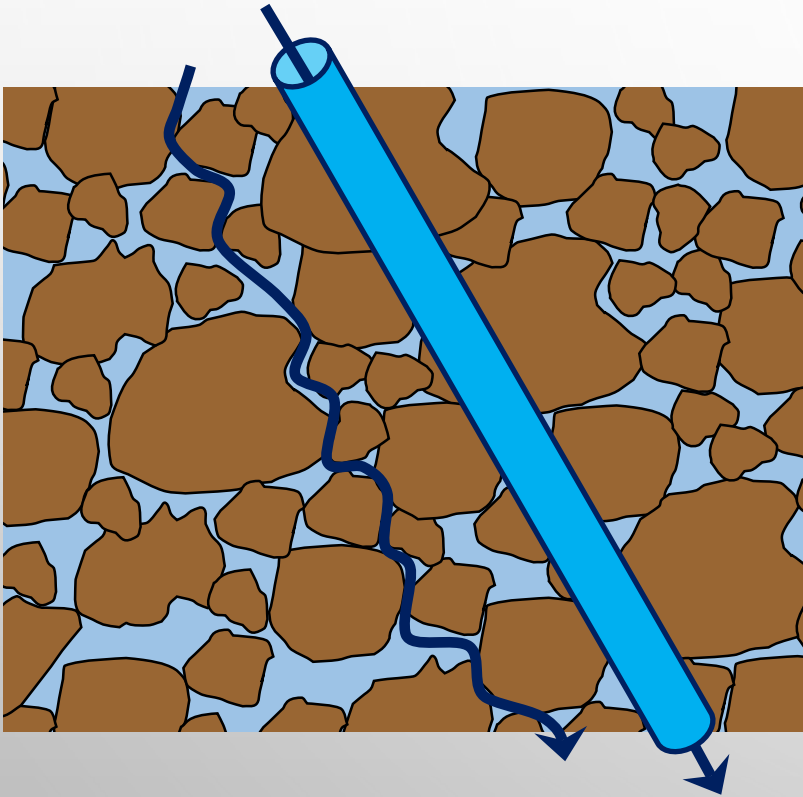


- Darcy's law for flow through a porous medium:

$$q = -K \frac{dh}{dl}$$

q = specific discharge Q/A [L/T]

Water movement through soil – saturated conditions

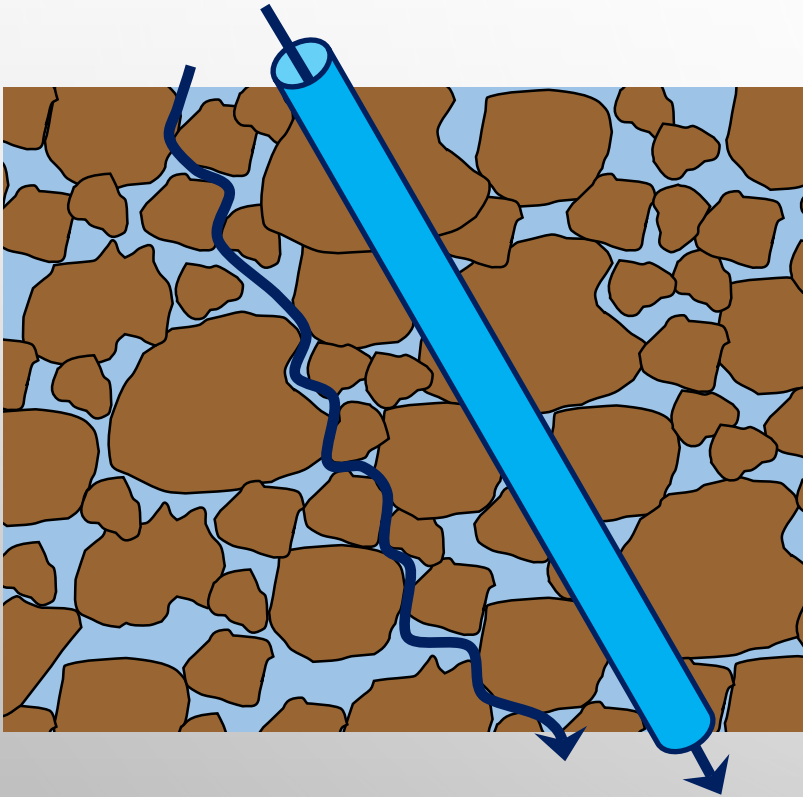


- Darcy's law for flow through a porous medium:

$$q = -K \frac{dh}{dl}$$

- K = hydraulic conductivity [L/T]

Water movement through soil – saturated conditions

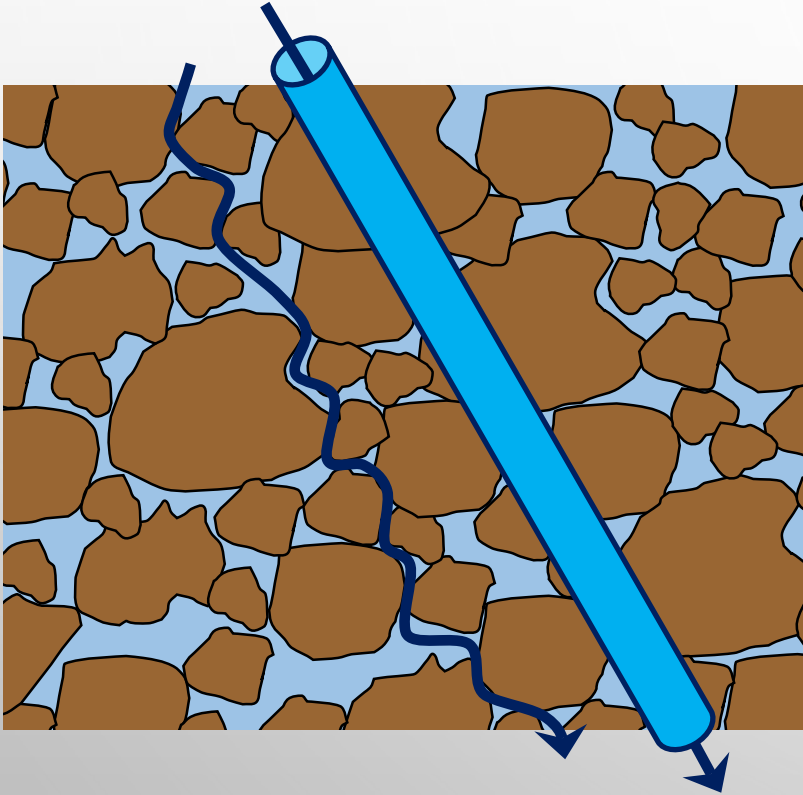


- Darcy's law for flow through a porous medium:

$$q = -K \frac{dh}{dl}$$

- K = hydraulic conductivity [L/T]
- K is a constant under saturated conditions (depends on properties of medium and fluid)

Water movement through soil – saturated conditions

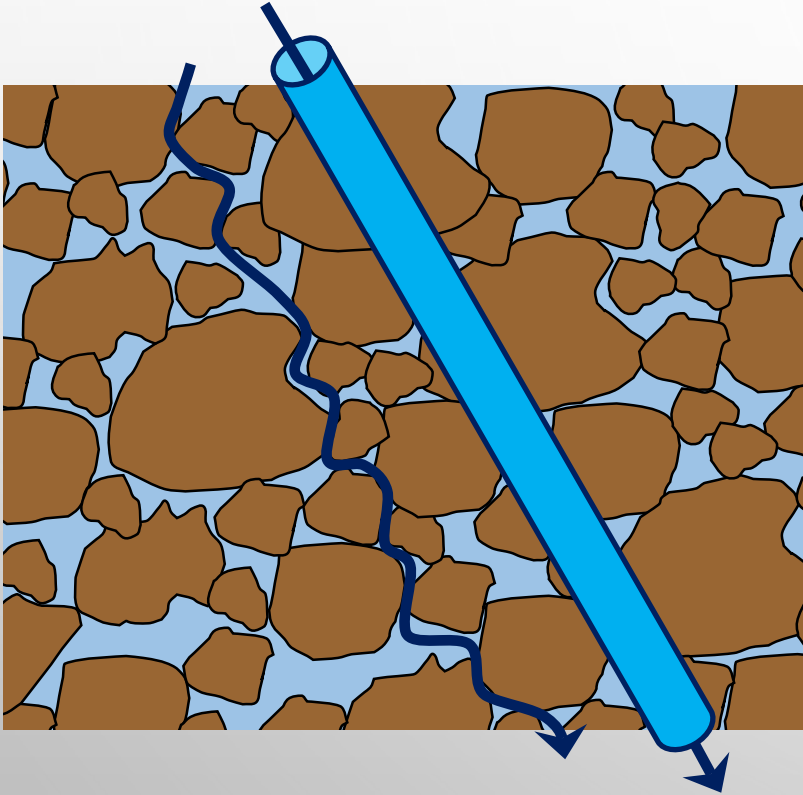


- Darcy's law for flow through a porous medium:

$$q = -K \frac{dh}{dl}$$

- h = hydraulic head [L], measure of the mechanical energy per unit weight of water

Water movement through soil – saturated conditions

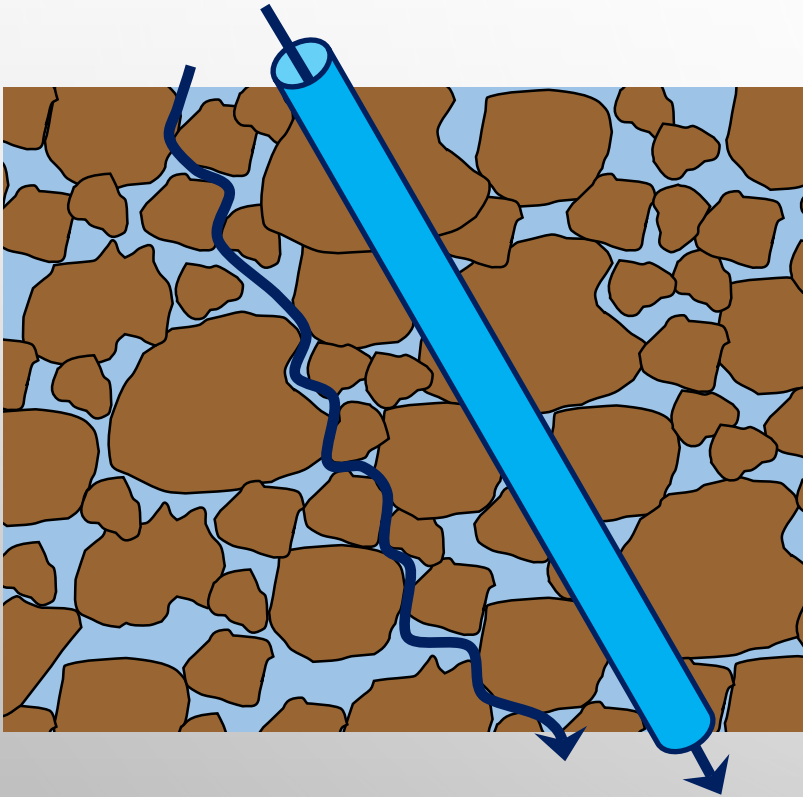


- Darcy's law for flow through a porous medium:

$$q = -K \frac{dh}{dl}$$

- h = hydraulic head [L], measure of the mechanical energy per unit weight of water
- Typically, three components: velocity head, elevation head, pressure head

Water movement through soil – saturated conditions

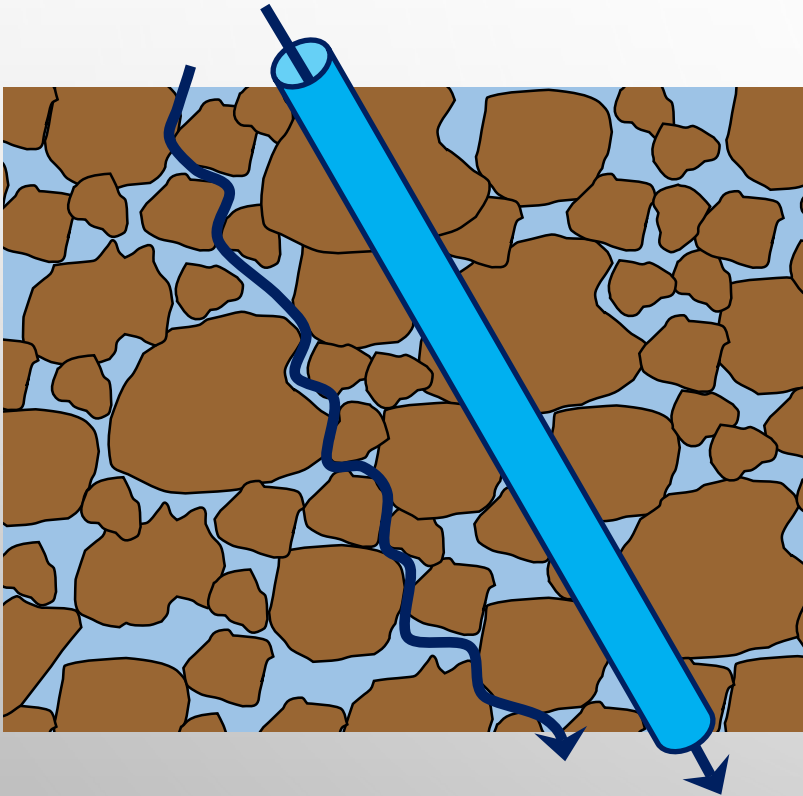


- Darcy's law for flow through a porous medium:

$$q = -K \frac{dh}{dl}$$

- h = hydraulic head [L], measure of the mechanical energy per unit weight of water
- Typically, three components: velocity head, elevation head, pressure head
- Based on analysis of forces acting on a fluid parcel

Water movement through soil – saturated conditions



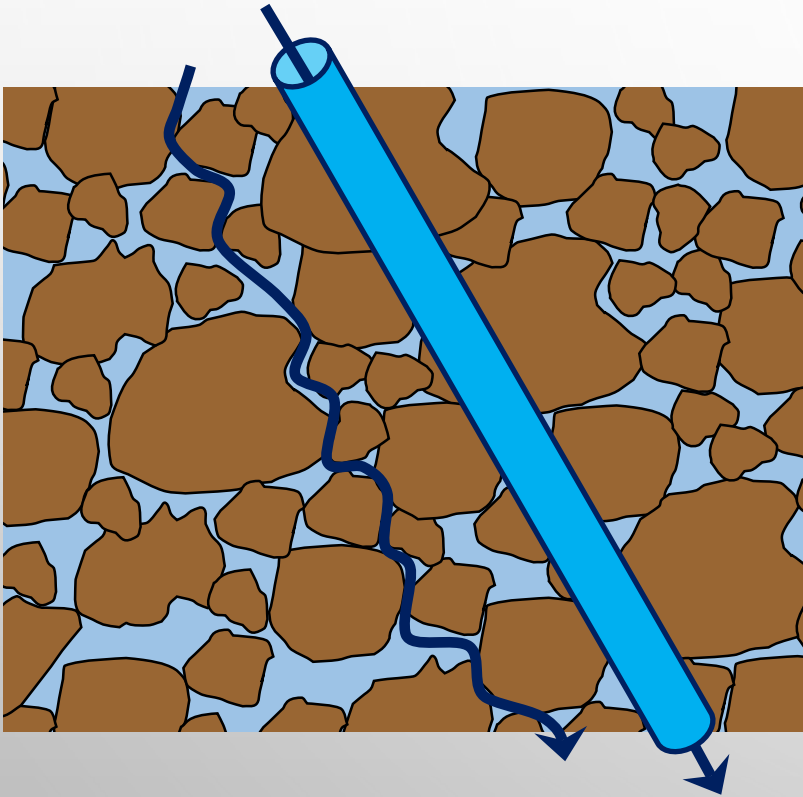
Low velocities in GW flow

- Darcy's law for flow through a porous medium:

$$q = -K \frac{dh}{dl}$$

- h = hydraulic head [L], measure of the mechanical energy per unit weight of water
- Typically, three components:
~~velocity head~~, elevation head, pressure head

Water movement through soil – saturated conditions

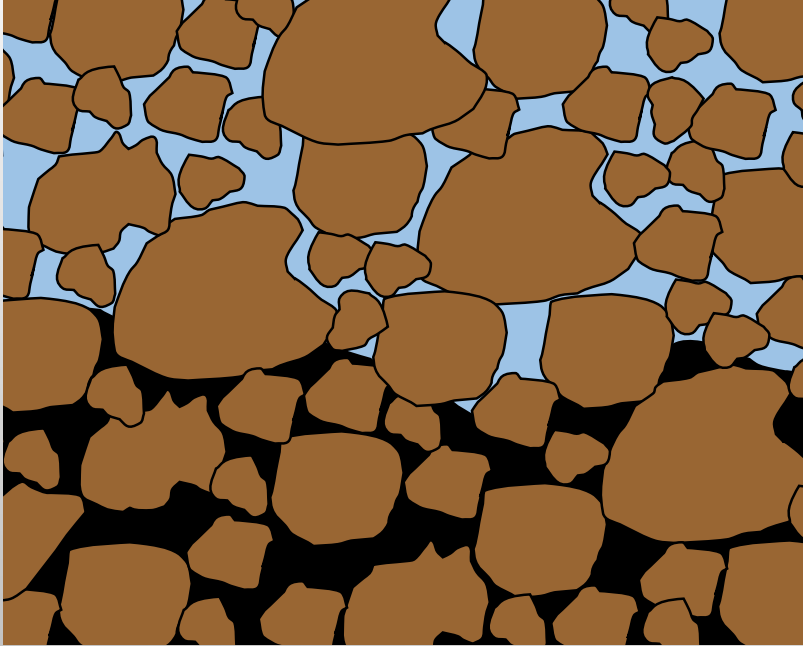


- Darcy's law for flow through a porous medium:

$$q = -k \frac{dh}{dl}$$

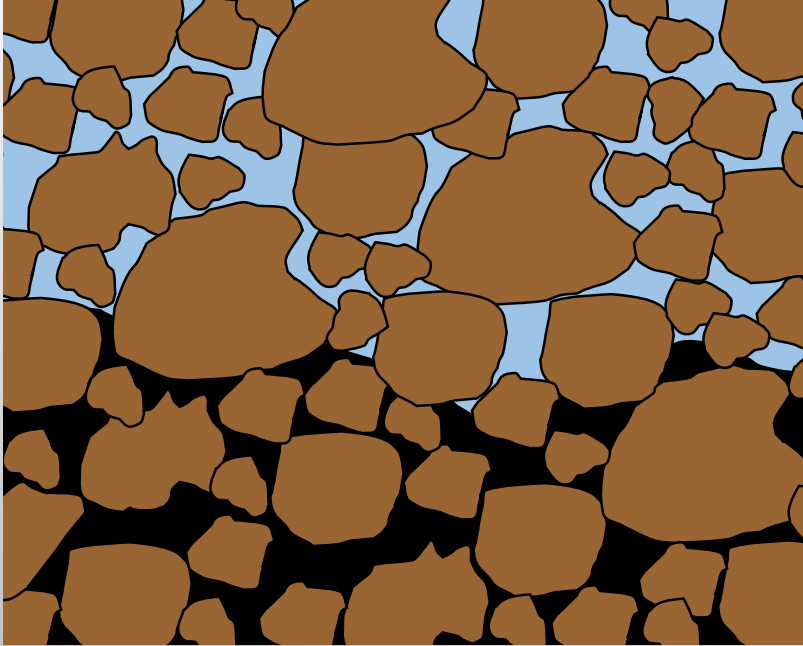
- l = length [L]

Water movement through soil – unsaturated conditions



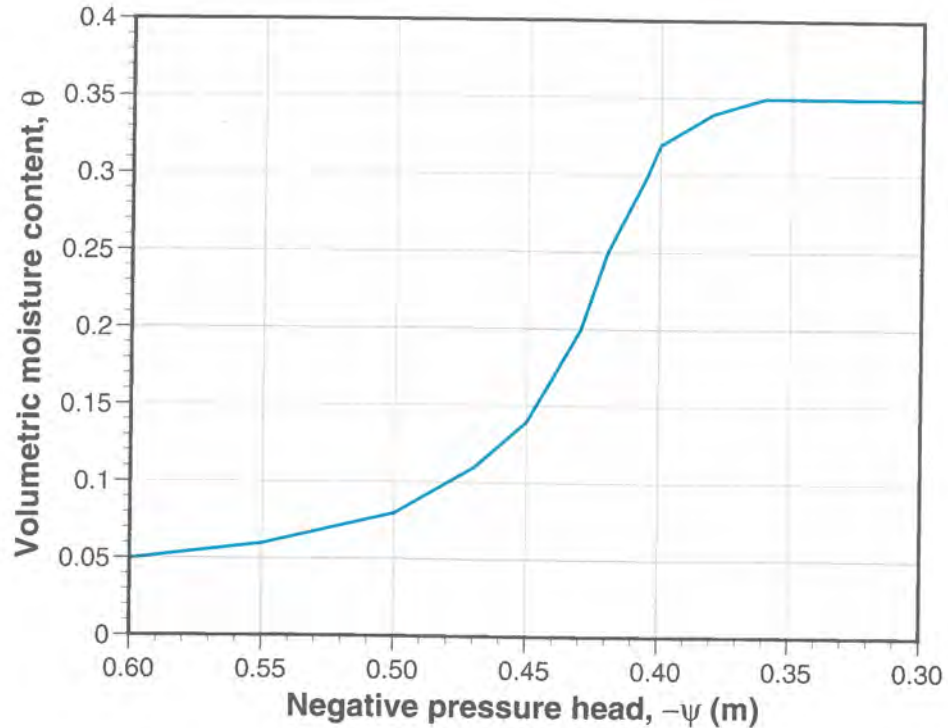
- Three-phase system of soil, water and air

Water movement through soil – unsaturated conditions



- Three-phase system of soil, water and air
- Things get more complicated:
 - Capillary forces cause negative pressure head (suction) ψ

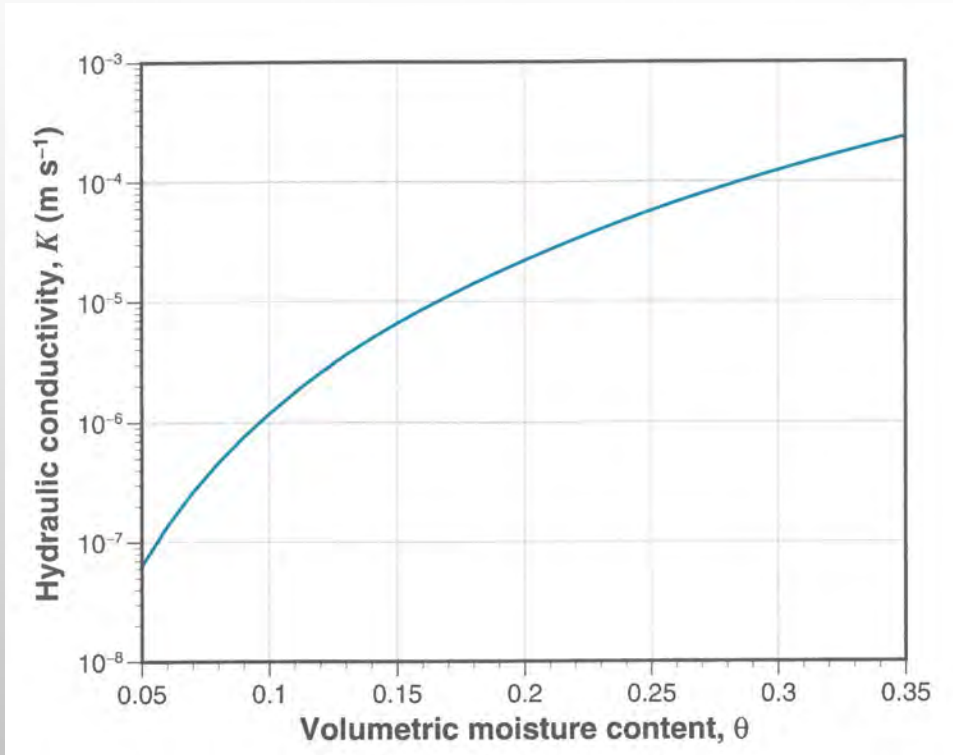
Water movement through soil – unsaturated conditions



- Three-phase system of soil, water and air
- Things get more complicated:
 - Capillary forces cause negative pressure head (suction) ψ
 - ψ varies with moisture content θ

Source: Hornberger et al. (2014). Elements of Physical Hydrology.

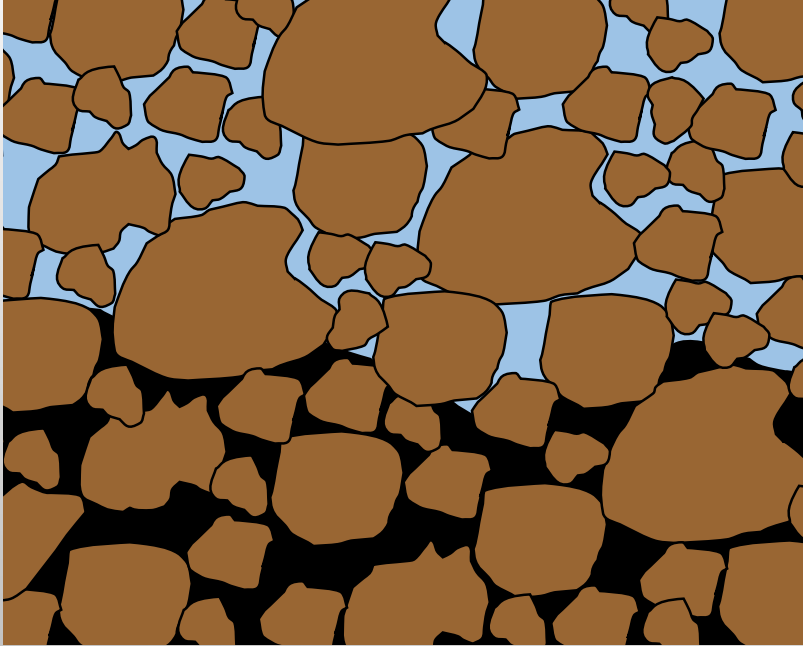
Water movement through soil – unsaturated conditions



- Three-phase system of soil, water and air
- Things get more complicated:
 - Capillary forces cause negative pressure head (suction) ψ
 - Hydraulic conductivity is not a constant, but varies with moisture content

Source: Hornberger et al. (2014). Elements of Physical Hydrology.

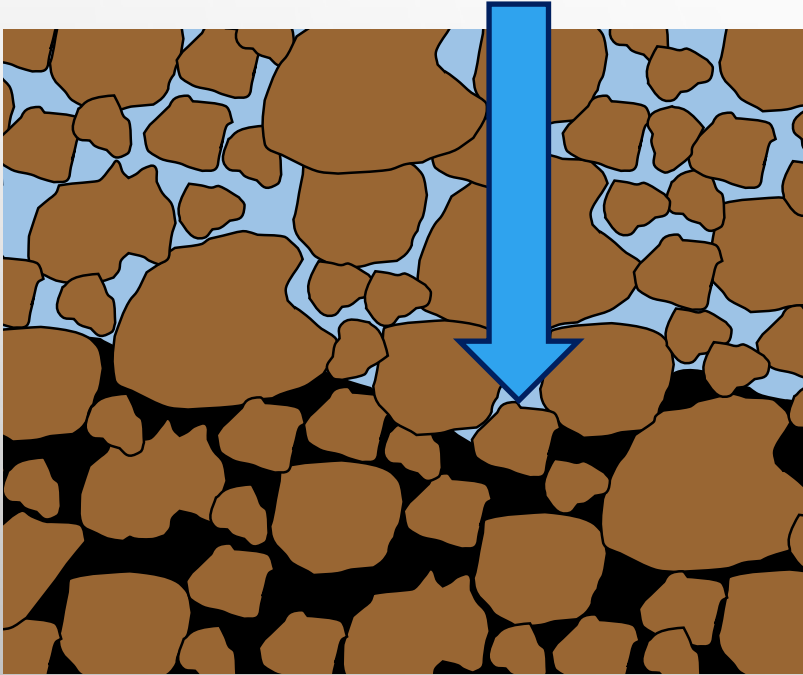
Water movement through soil – unsaturated conditions



- Darcy's law for unsaturated conditions
(steady flow of soil moisture)

$$q = -K(\theta) \frac{dh}{dl}$$

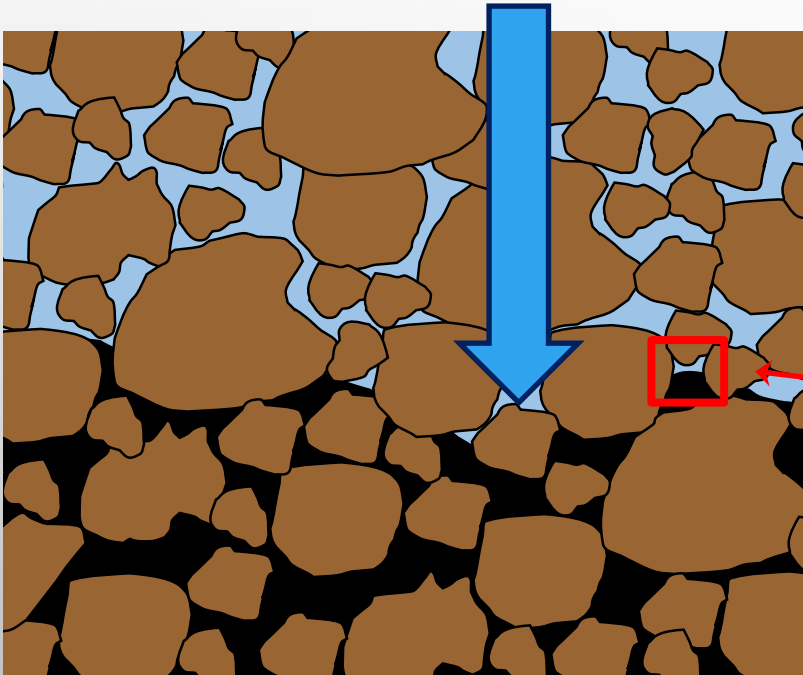
Water movement through soil – unsaturated conditions



- Darcy's law for unsaturated conditions (steady flow of soil moisture)
- For one-dimensional flow in the vertical direction:

$$q_z = -K(\theta) \frac{dh}{dz}$$

Water movement through soil – unsaturated conditions

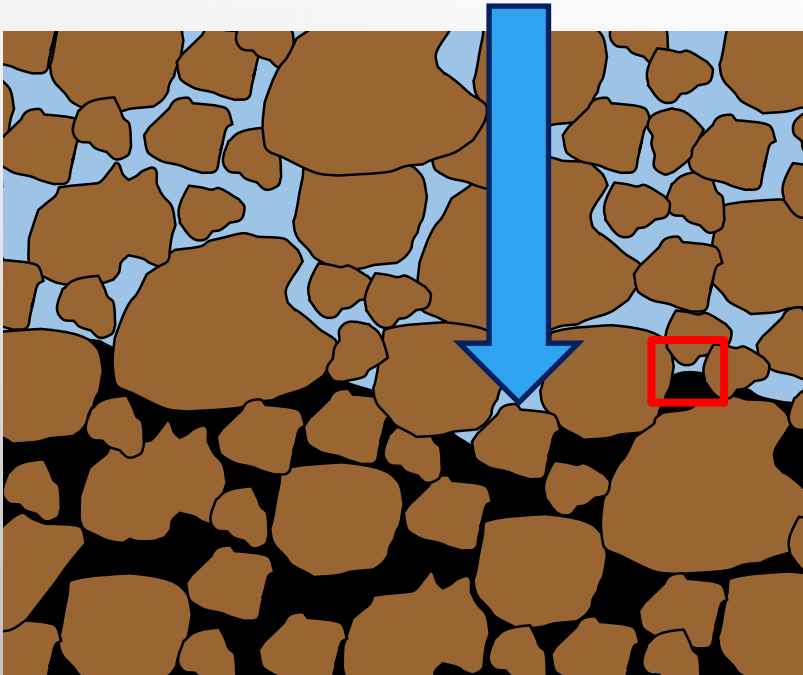


Small control
volume

- But it gets more complicated: infiltration is an unsteady (transient) process; soil moisture changes with time:

$$\underbrace{\frac{d\theta}{dt}}_{\text{Change in soil moisture with time}} = \underbrace{-\frac{dq_z}{dz}}_{\text{Difference between inflow and outflow}} \quad \text{Continuity equation}$$

Water movement through soil – unsaturated conditions

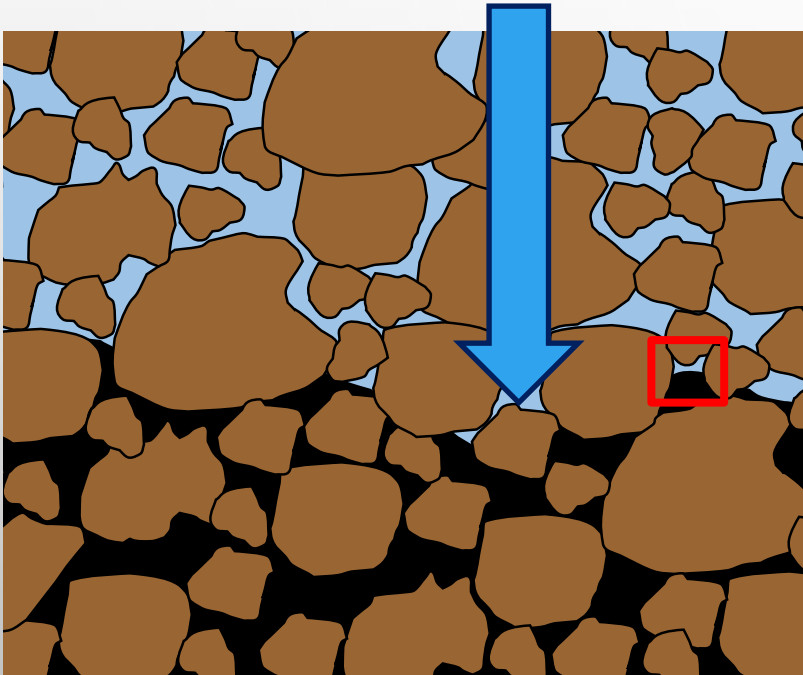


Small
control
volume

- Combine with Darcy's law:

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left[K(\theta) \frac{\partial h}{\partial z} \right]$$

Water movement through soil – unsaturated conditions



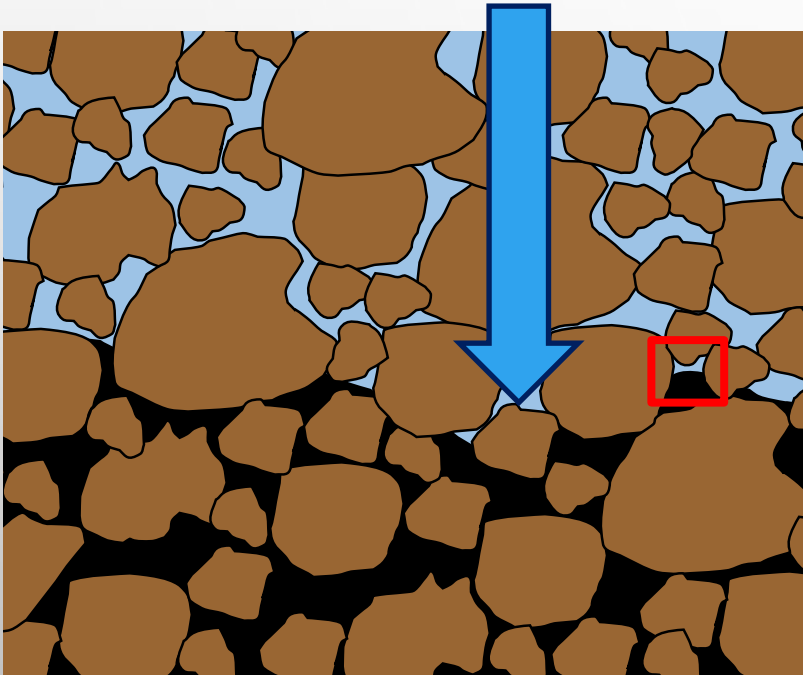
Small
control
volume

- Combine with Darcy's law:

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left[K(\theta) \frac{\partial (\psi + z)}{\partial z} \right]$$

$h = \psi + z$

Water movement through soil – unsaturated conditions

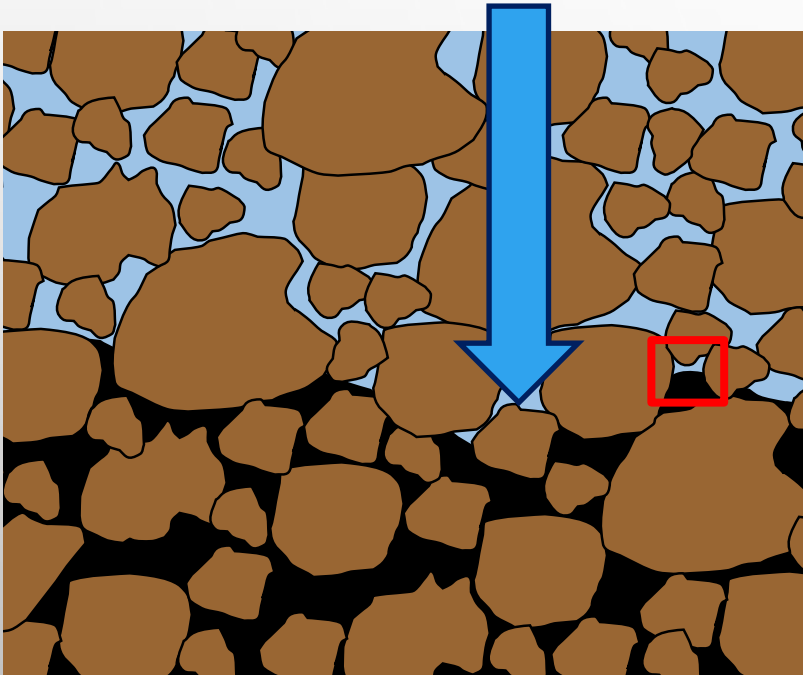


Small
control
volume

- Richard's equation for vertical, unsaturated, transient flow:

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left[K(\theta) \left(\frac{\partial \psi}{\partial z} + 1 \right) \right]$$

Water movement through soil – unsaturated conditions



Small
control
volume

- Richard's equation for vertical, unsaturated, transient flow:

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left[K(\theta) \left(\frac{\partial \psi}{\partial z} + 1 \right) \right]$$

How do we solve this?

Modeling infiltration

Physically-based model

- Start with physical laws
- Apply simplifying assumptions to solve Richard's equation

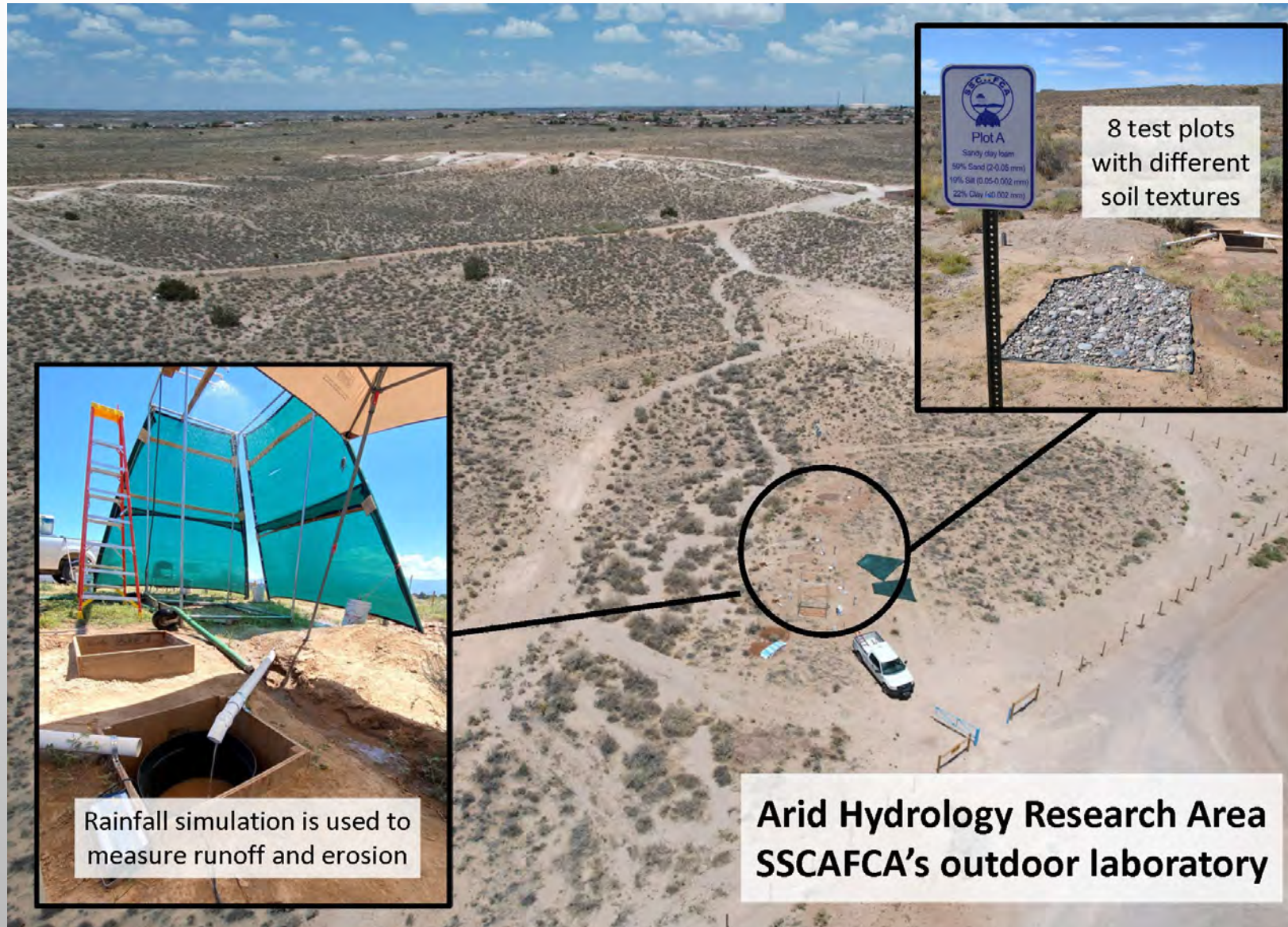
Conceptual model

- Start with conceptualization of core processes
- Find equations to describe processes

Empirical model

- Based on observations

What does a typical infiltration curve look like?

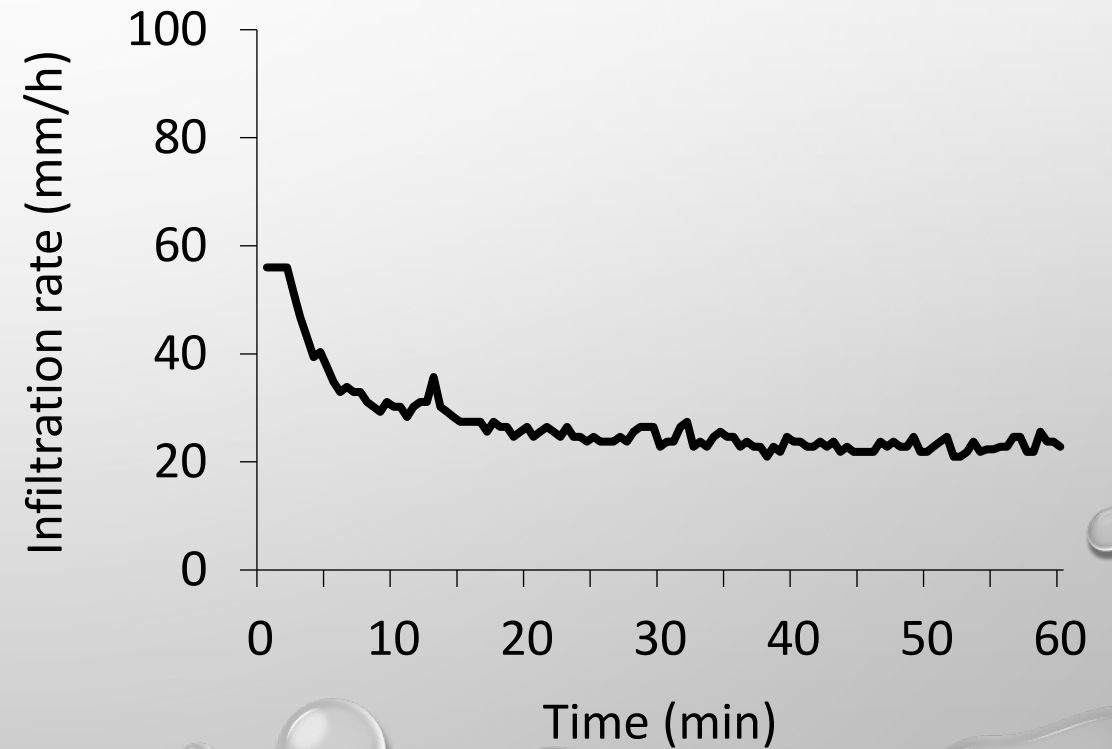




What does a typical infiltration curve look like?



More than 150 individual test runs!



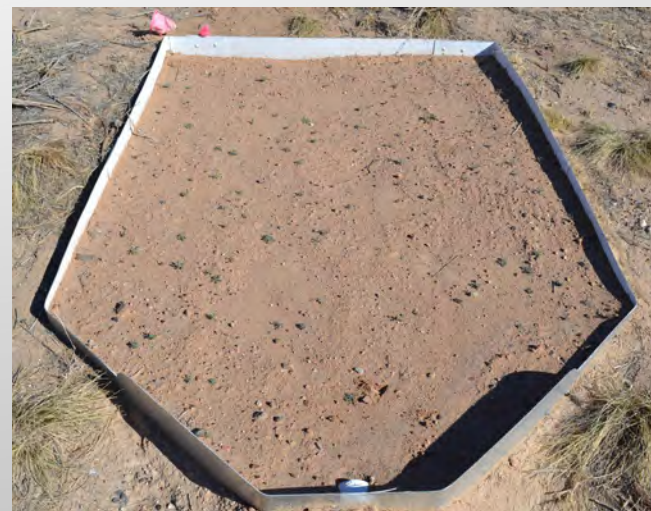
Test you intuition

Condition			Correct?
1	Uncompacted	Compacted	
2	Dry	Wet	
3	Crust	No crust	
4	Bare soil	Gravel mulch	
5	Plant canopy	Bare soil	
6	Fine	Coarse	
7	Steep	Less steep	
Number of correct answers:			

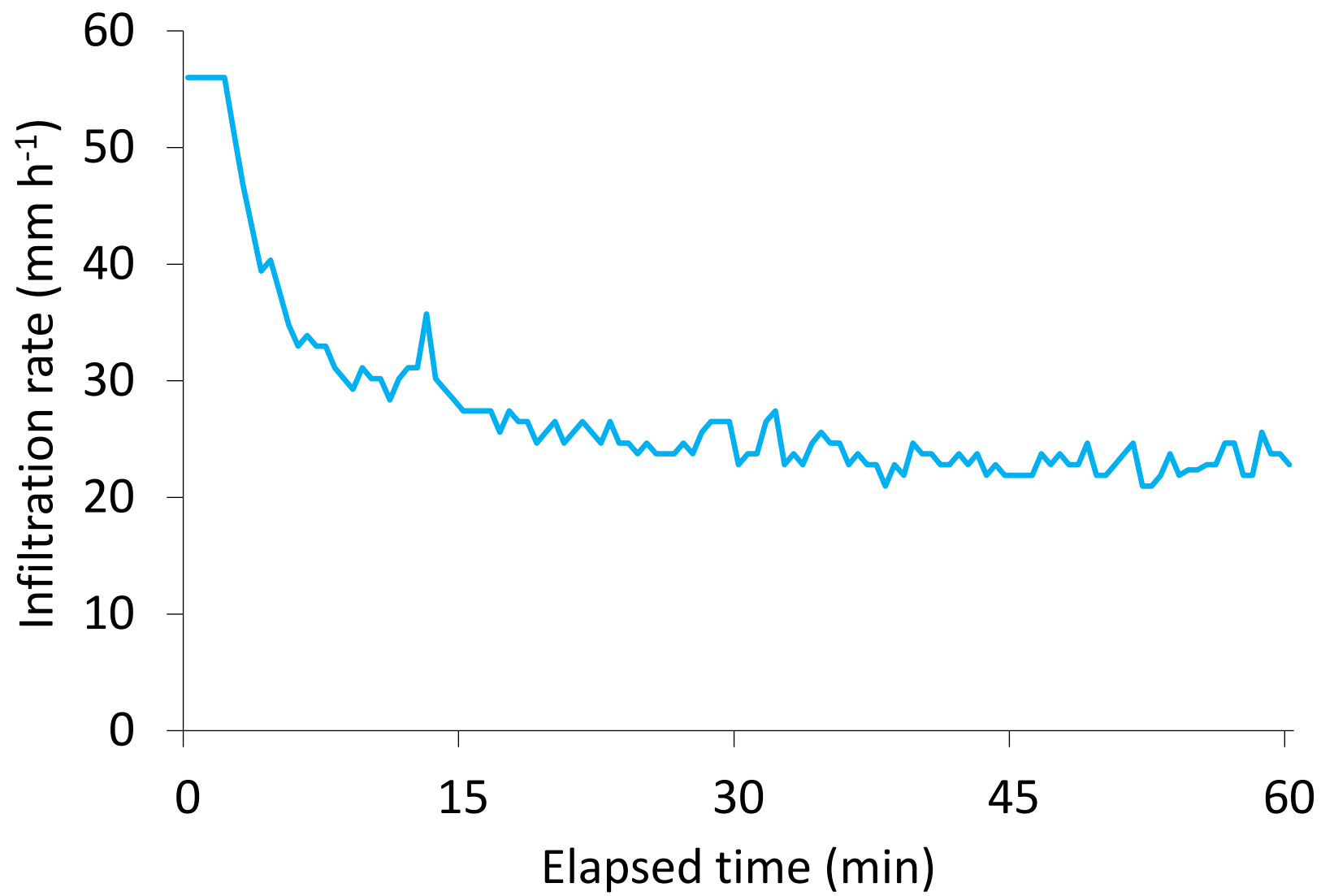
1. How does soil compaction impact infiltration?



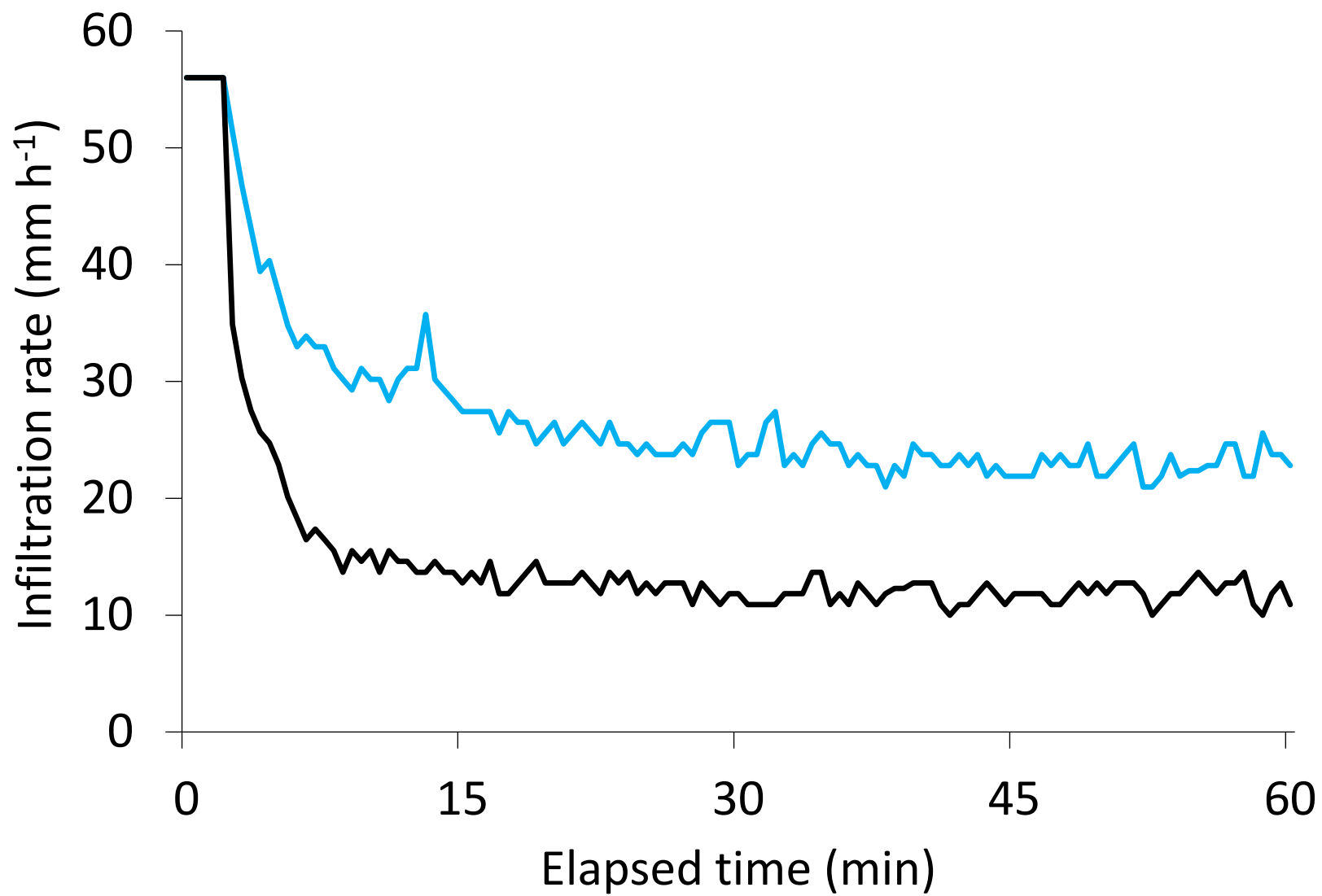
Compacted



**Un-
compacted**

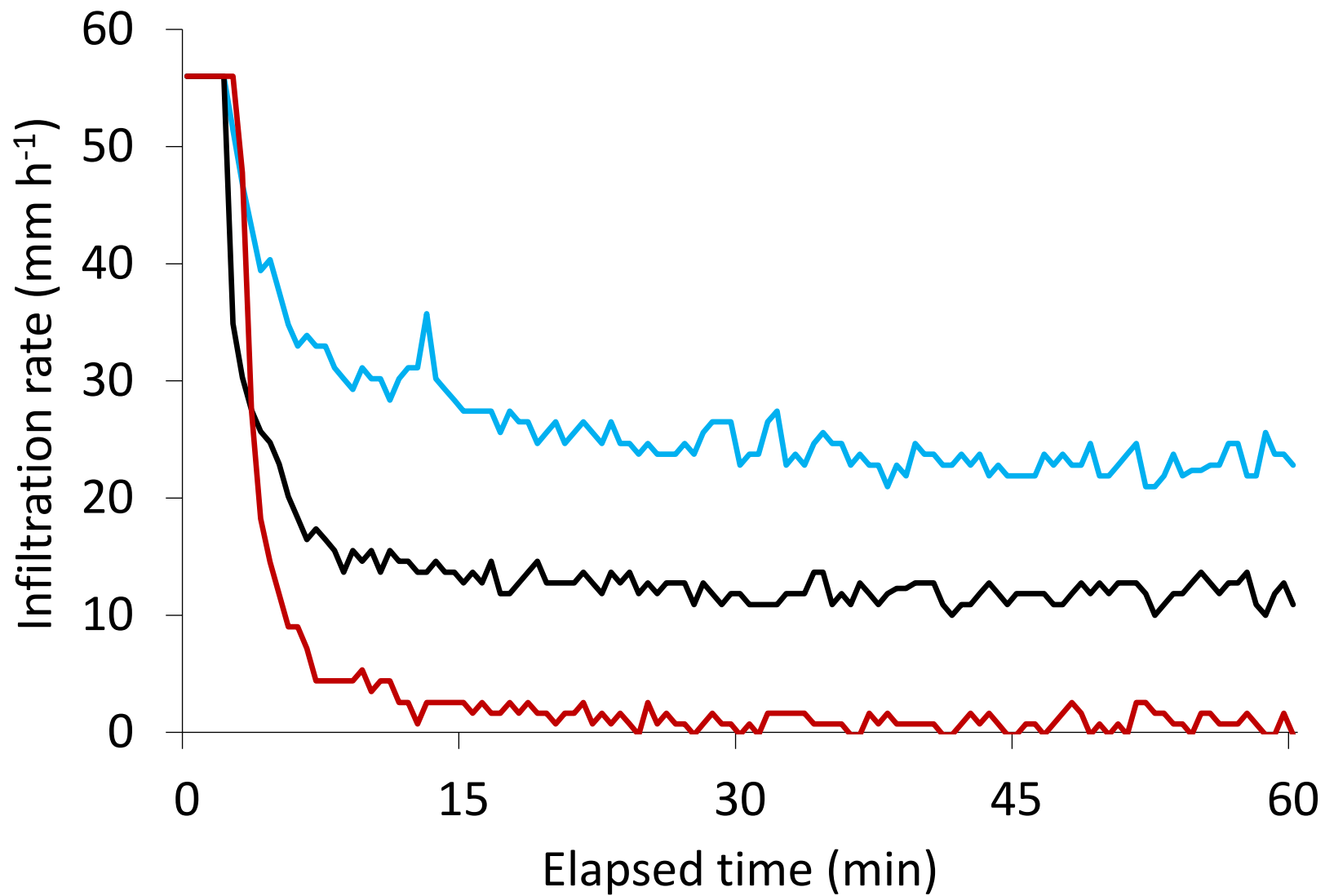


Uncompacted



Uncompacted

Incidental compaction



Uncompacted

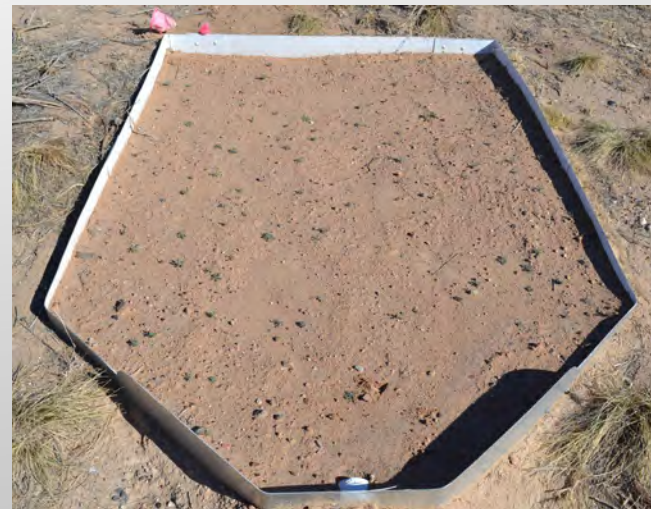
Incidental compaction

Mechanical compaction

1. How does soil compaction impact infiltration?



Compacted



**Un-
compacted**

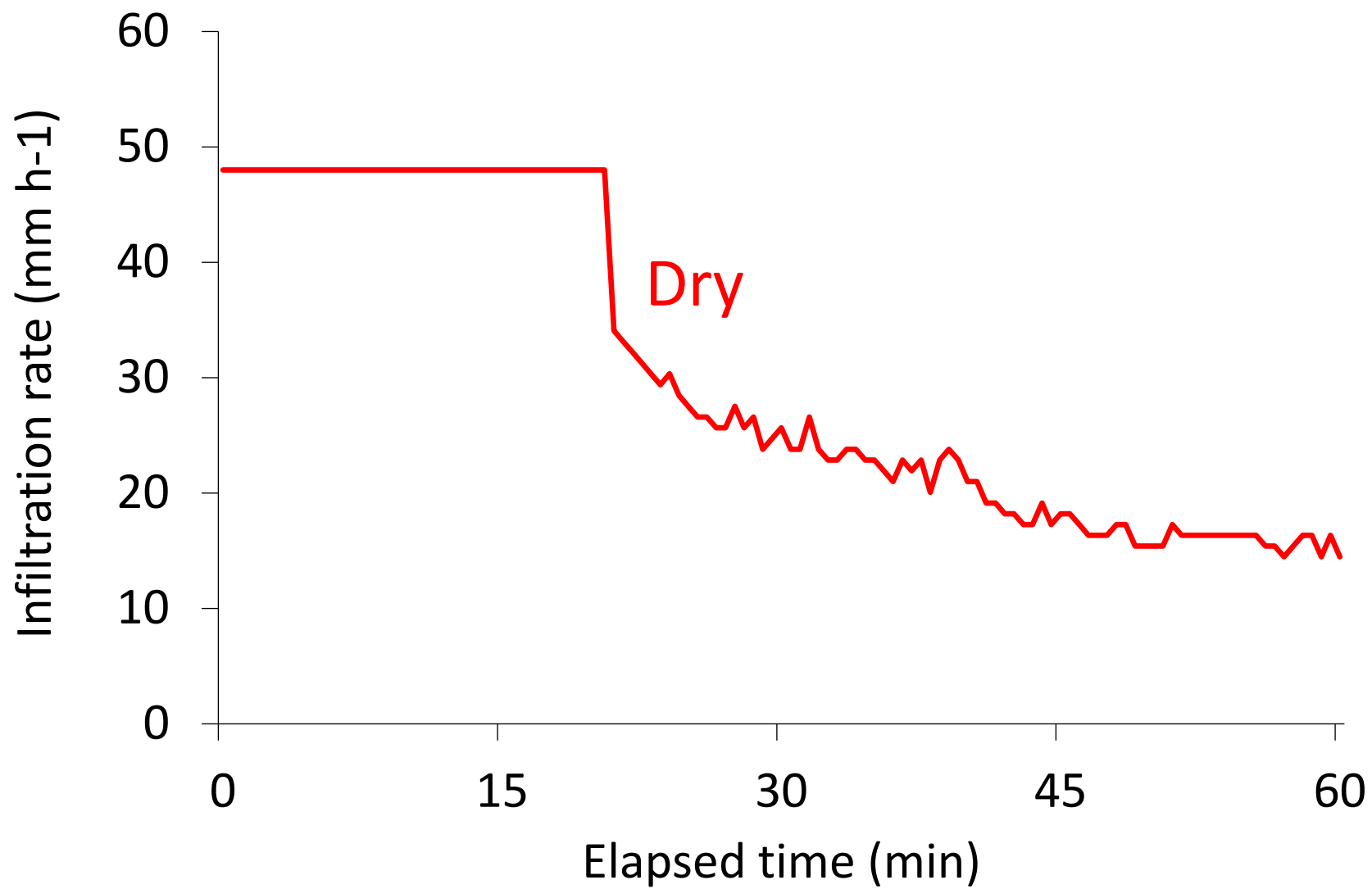
1. Compaction
2. Soil moisture

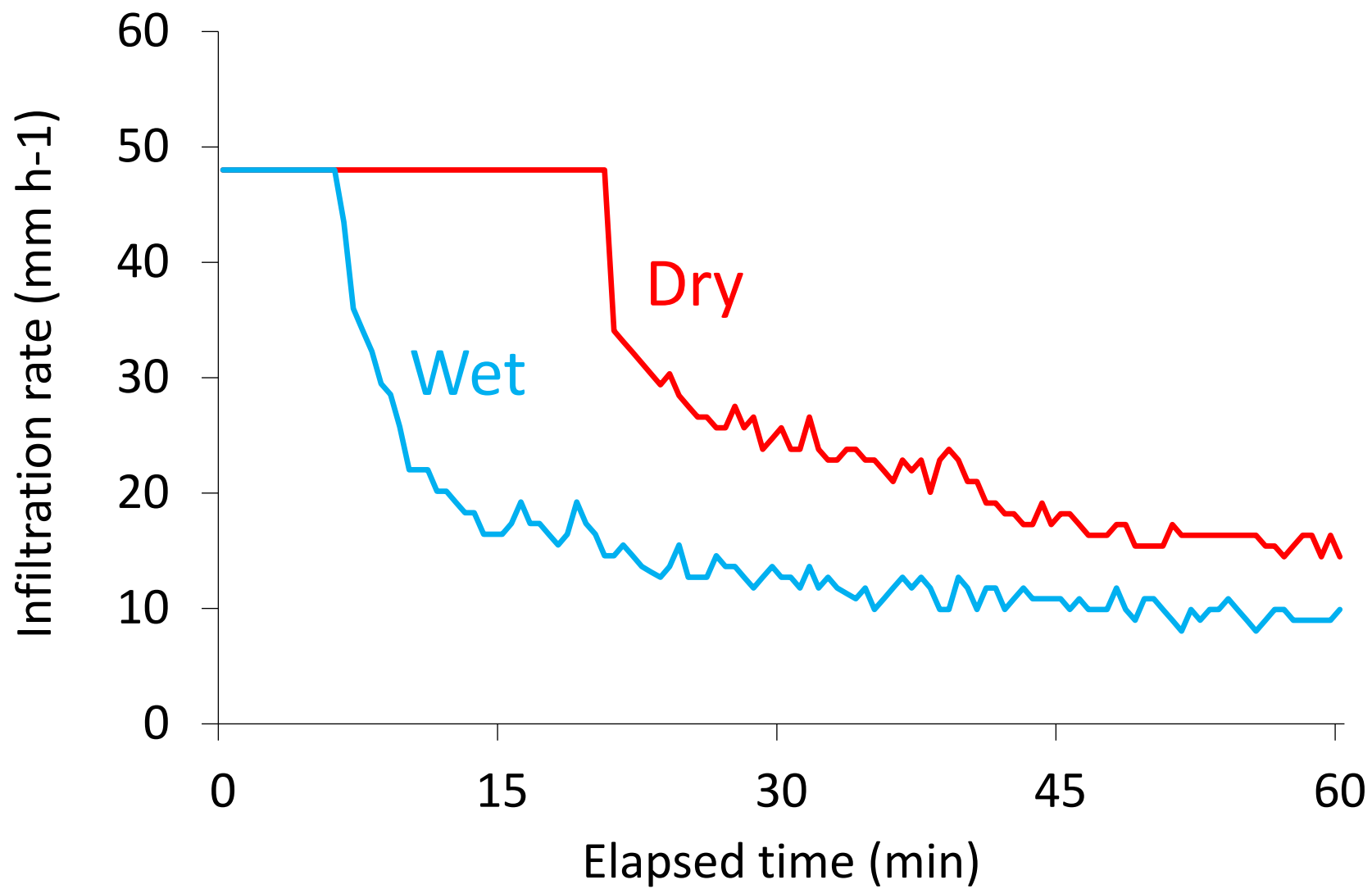


Dry



Wet





1. Compaction
2. Soil moisture



Dry



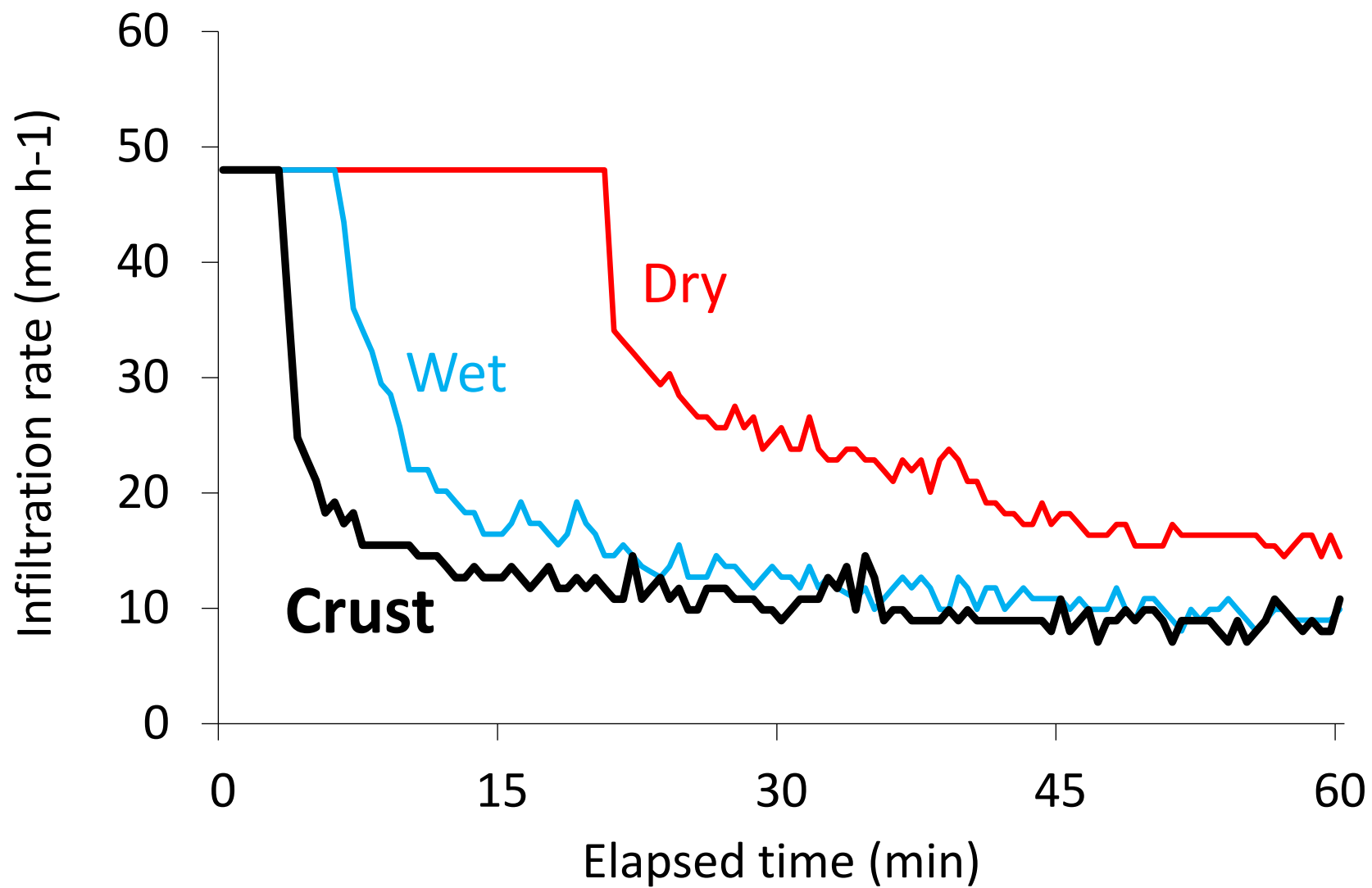
Wet

1. Compaction
2. Soil moisture
3. **Physical crust**



Crust

No crust



1. Compaction
2. Soil moisture
3. Physical crust



Crust

No crust

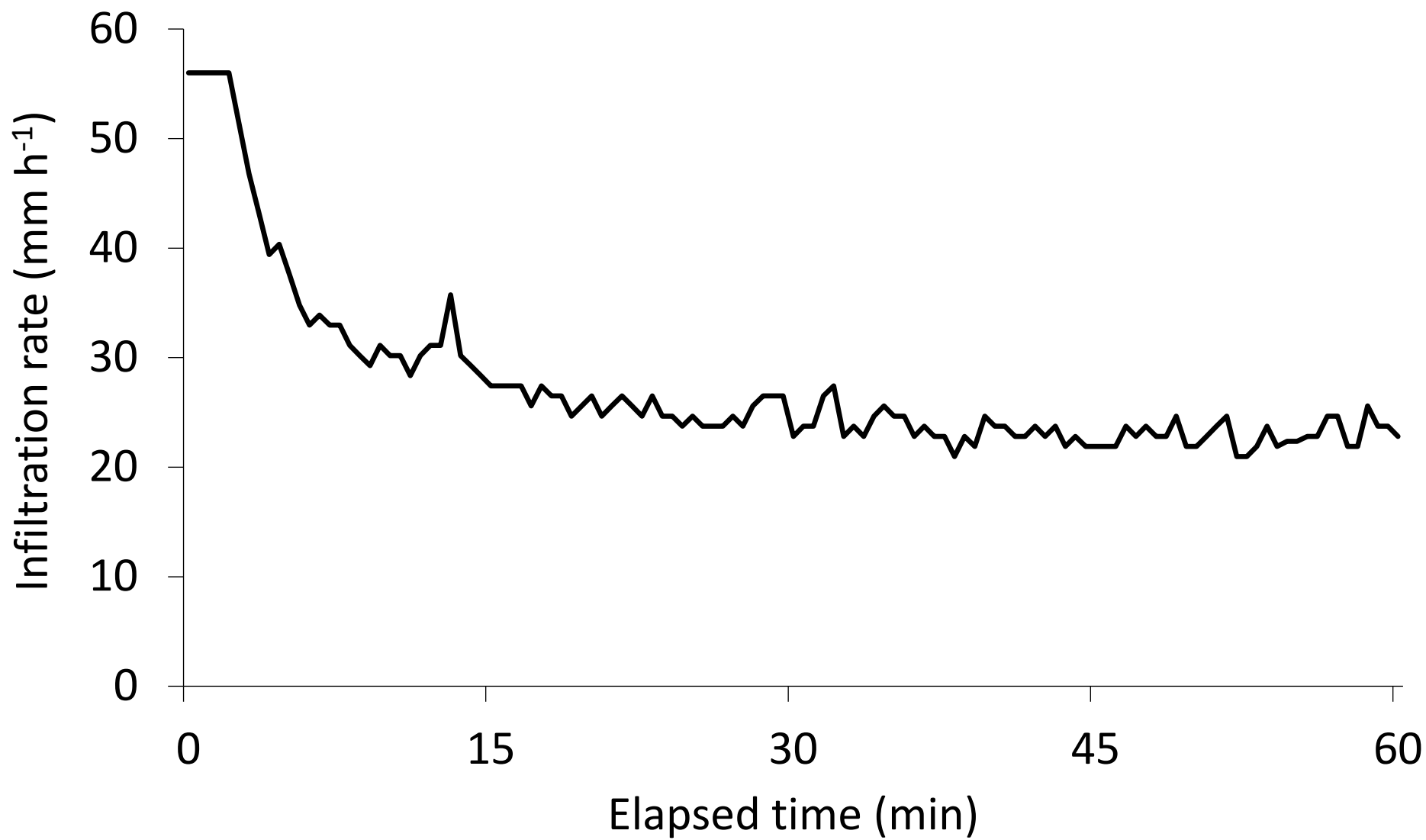
1. Compaction
2. Soil moisture
3. Physical crust
4. Gravel mulch



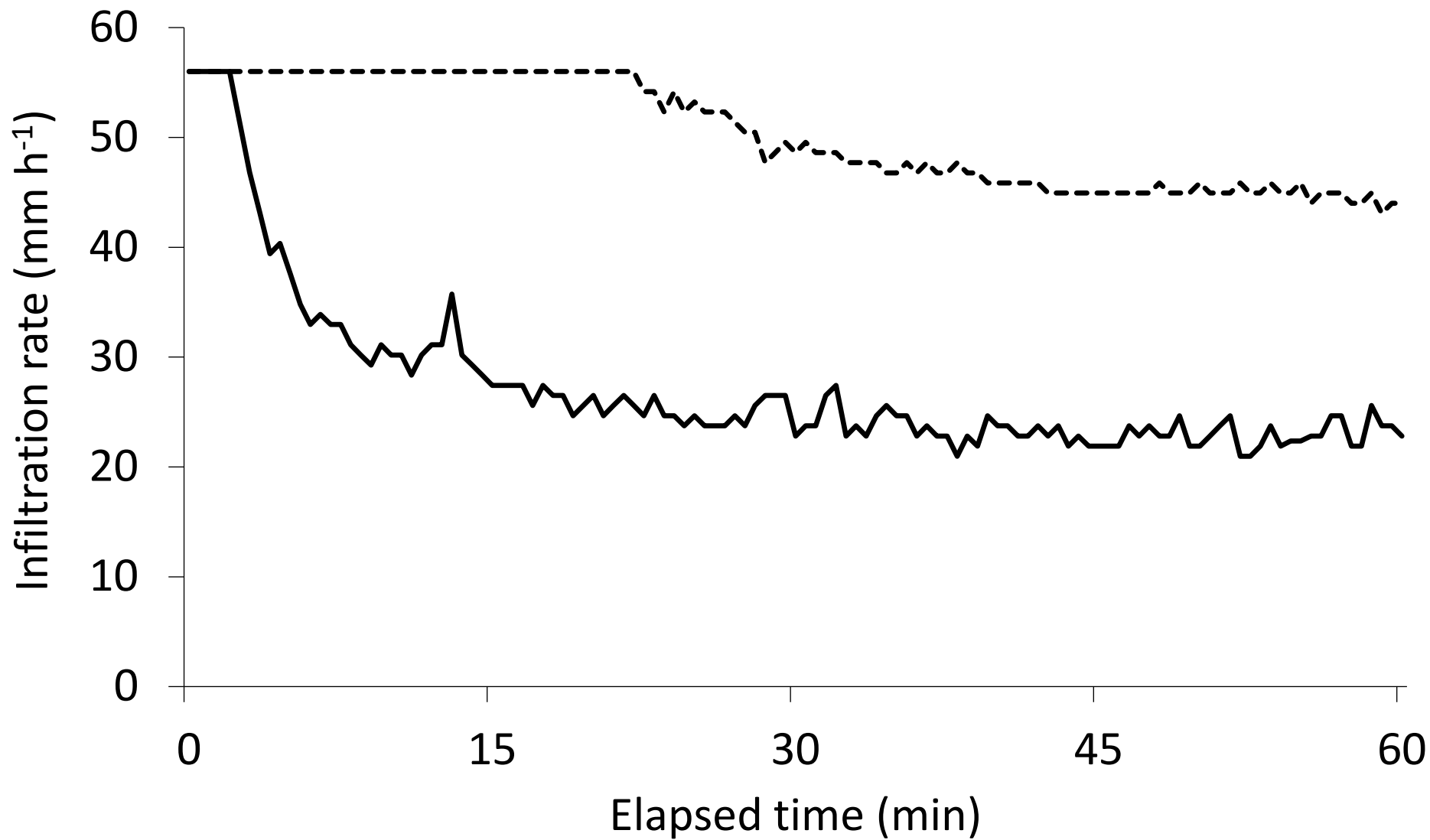
**Bare
soil**



**Gravel
mulch**



Bare soil



Gravel
mulch

Bare soil

1. Compaction
2. Soil moisture
3. Physical crust
4. Gravel mulch



**Bare
soil**

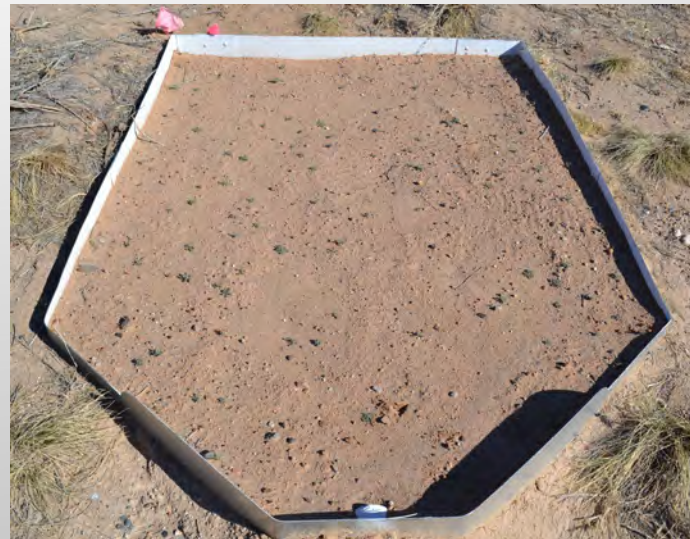


**Gravel
mulch**

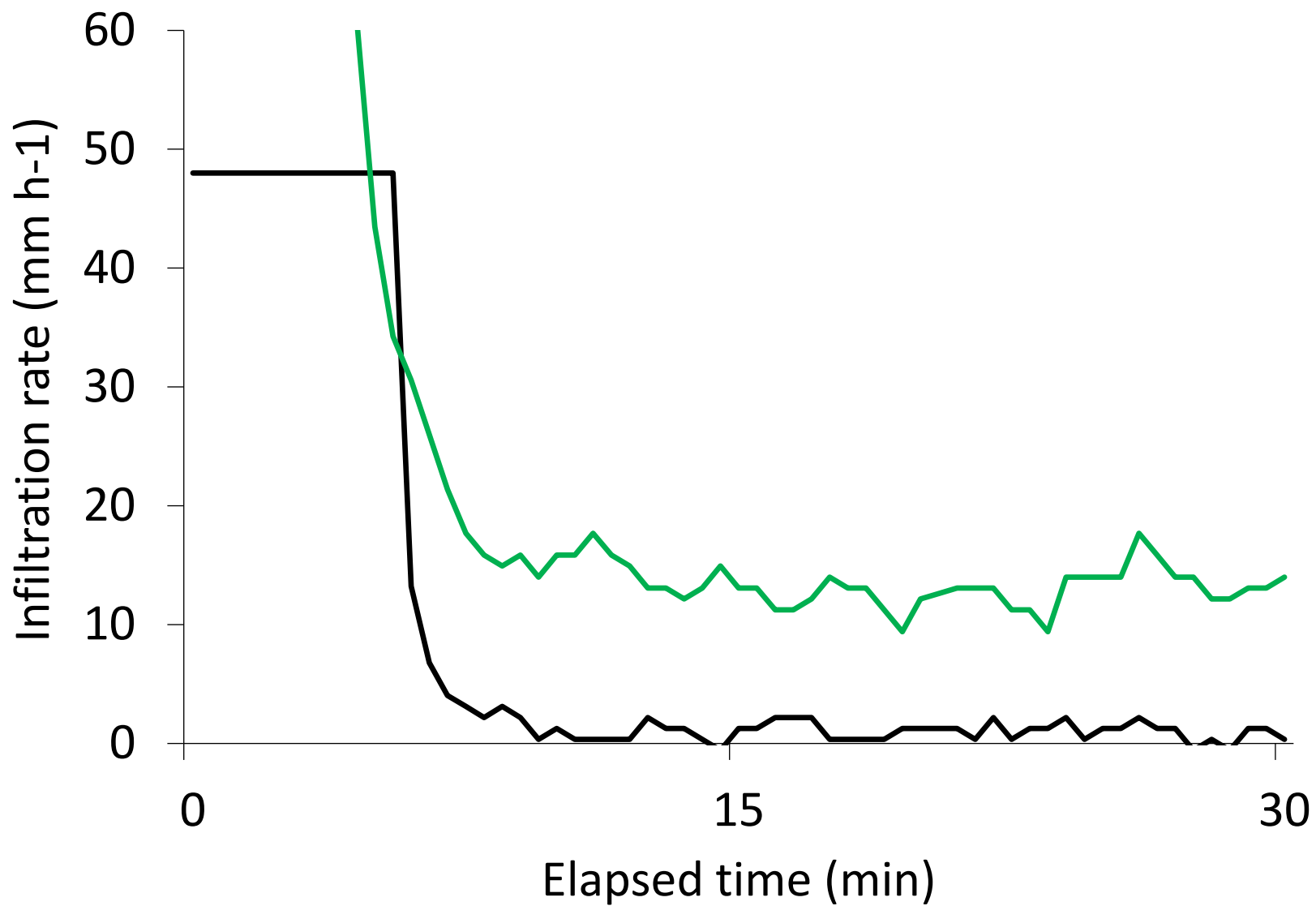
1. Compaction
2. Soil moisture
3. Physical crust
4. Gravel mulch
5. **Plant canopy**



**Plant
canopy**



**Bare
soil**



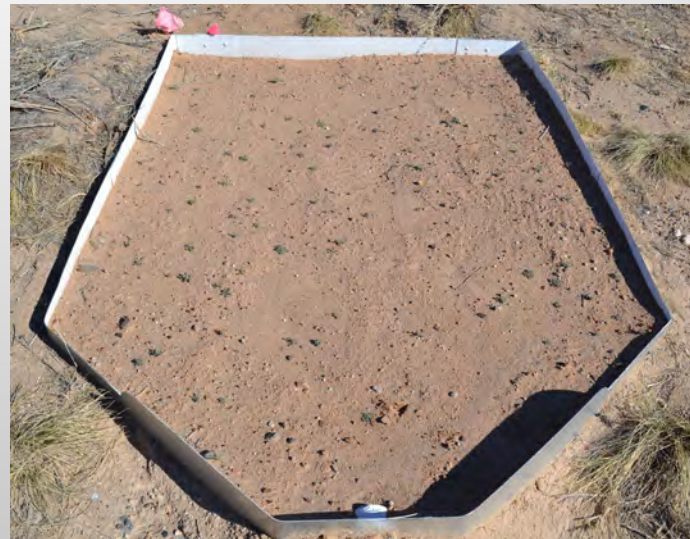
Plant Canopy

Bare soil

1. Compaction
2. Soil moisture
3. Physical crust
4. Gravel mulch
5. **Plant canopy**

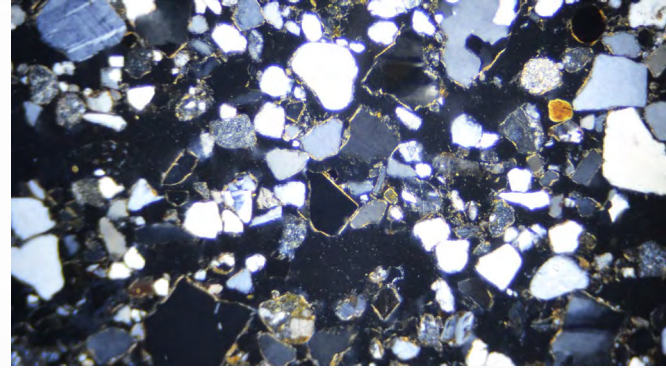


**Plant
canopy**

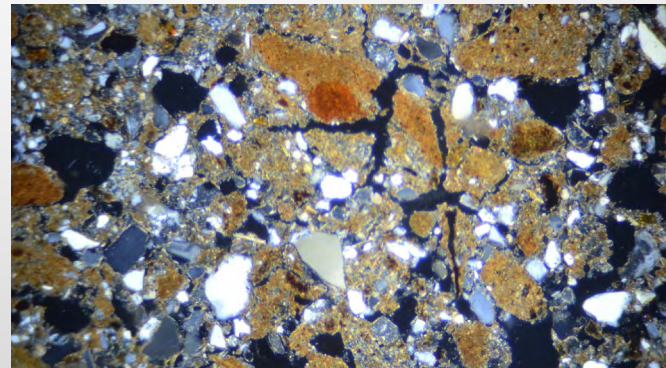


**Bare
soil**

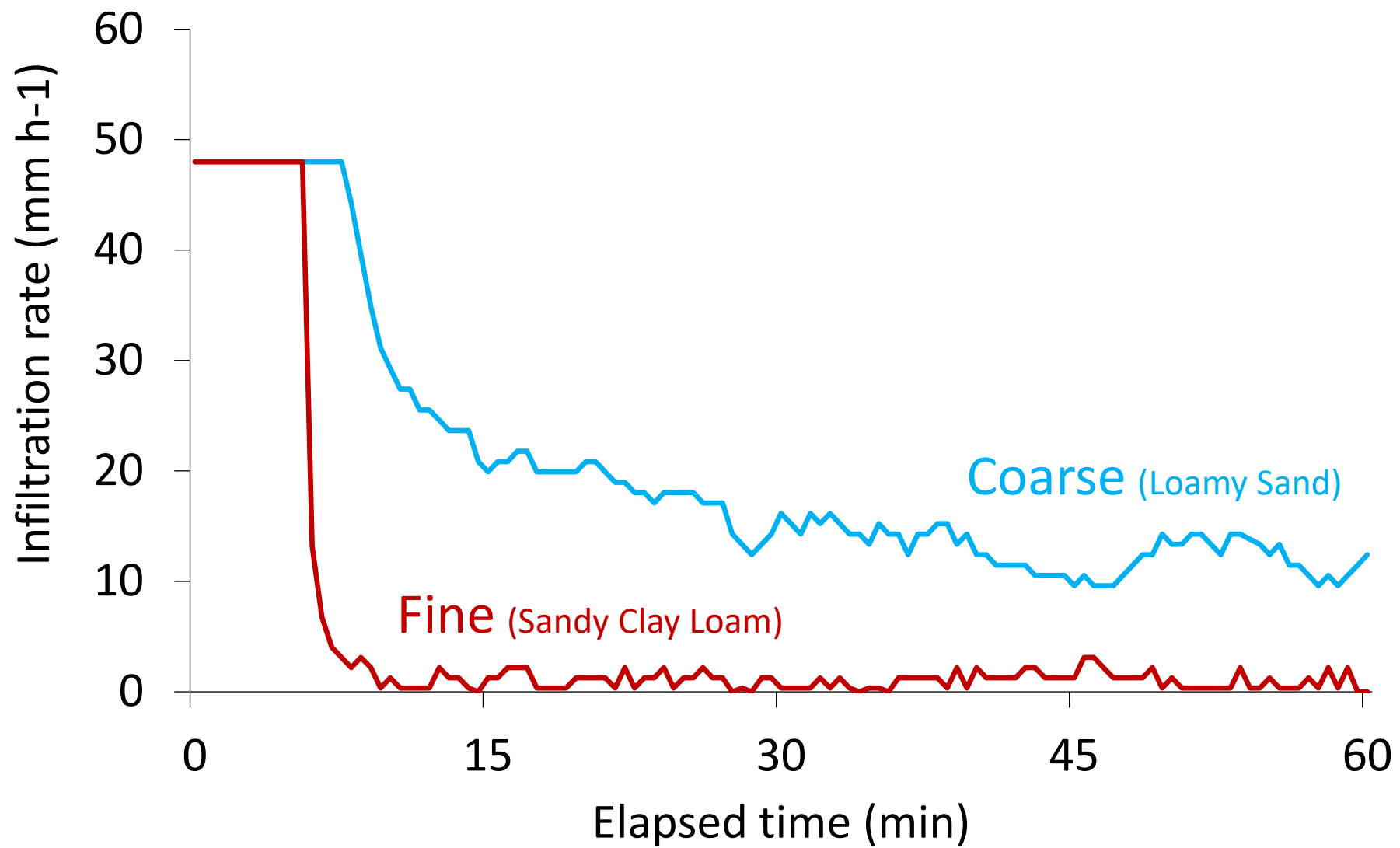
1. Compaction
2. Soil moisture
3. Physical crust
4. Gravel mulch
5. Plant canopy
6. **Soil texture**



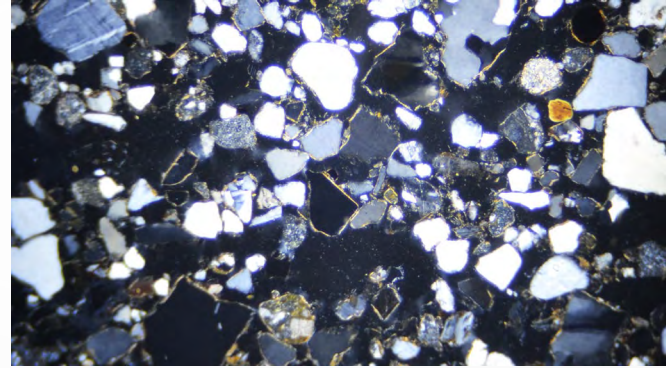
Coarse



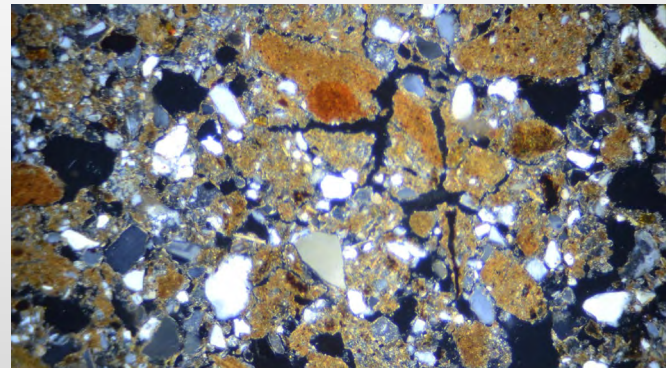
Fine



1. Compaction
2. Soil moisture
3. Physical crust
4. Gravel mulch
5. Plant canopy
6. **Soil texture**



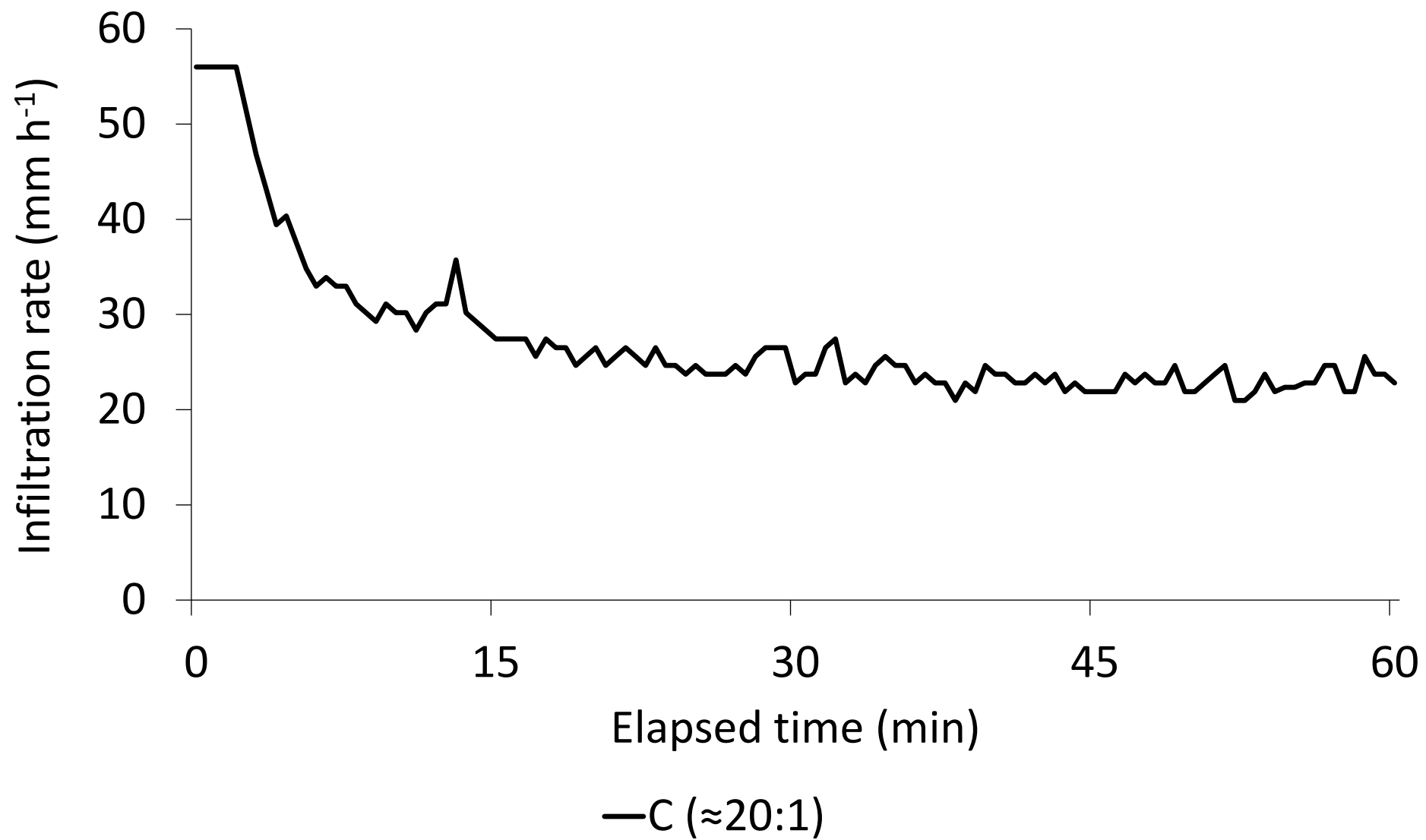
Coarse

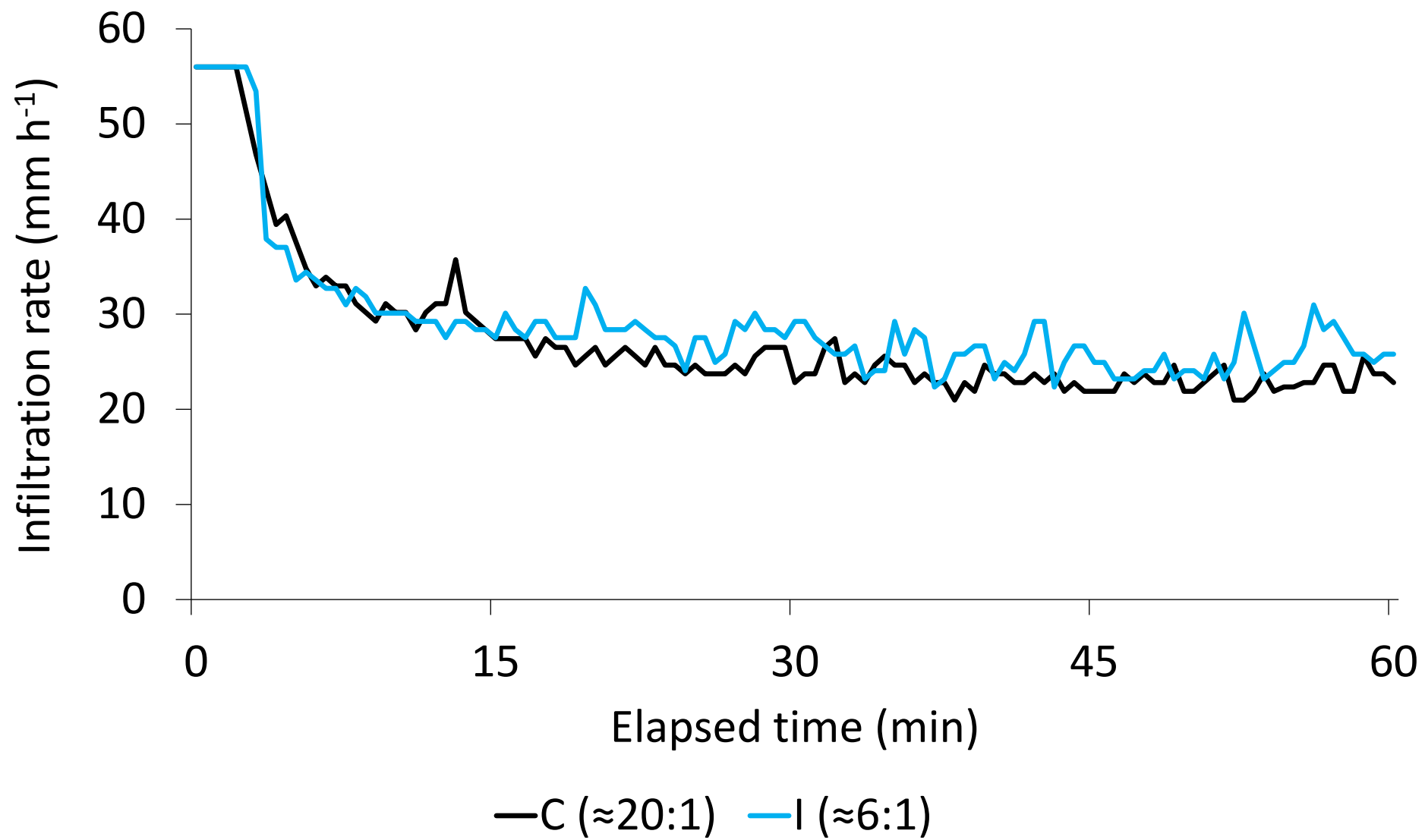


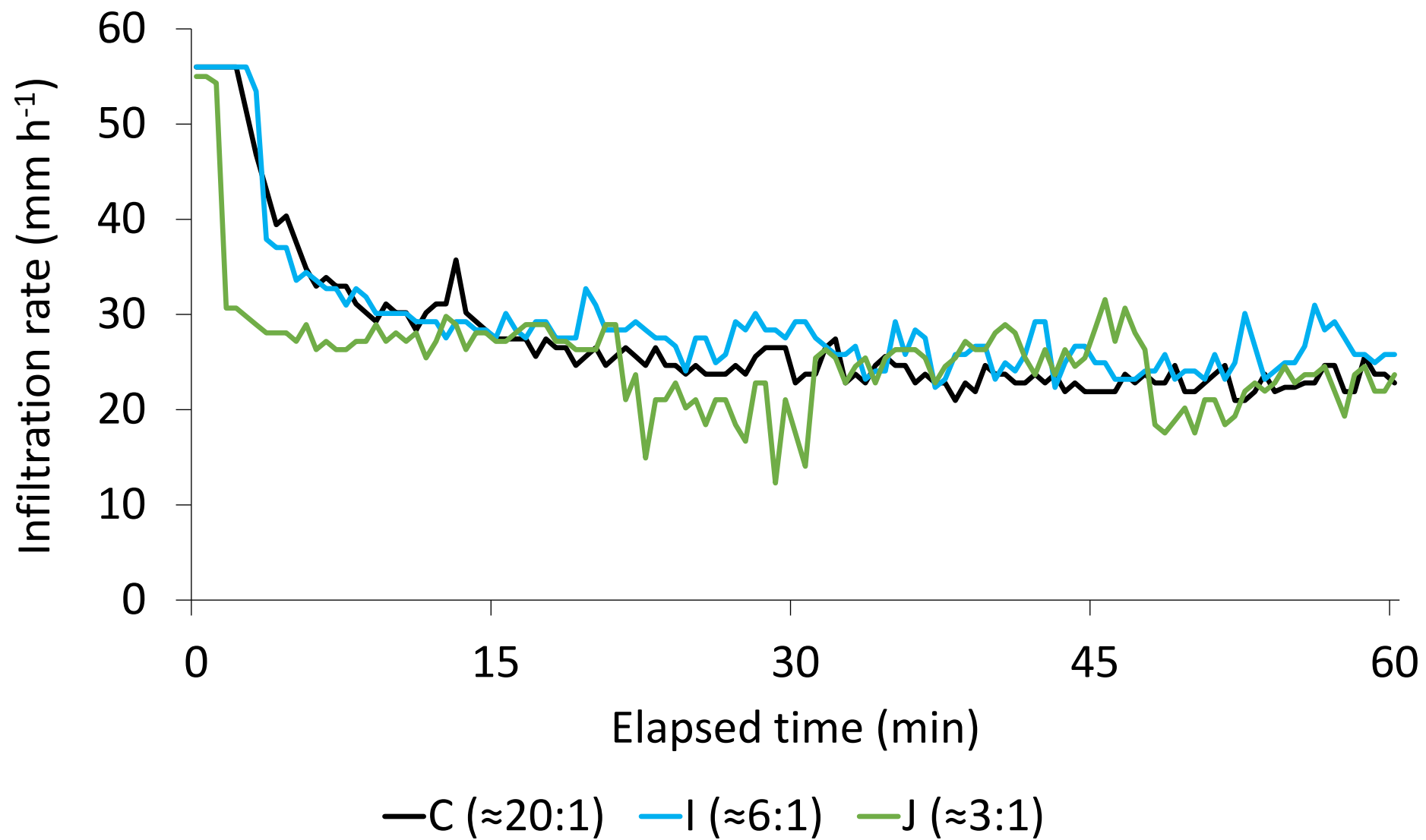
Fine

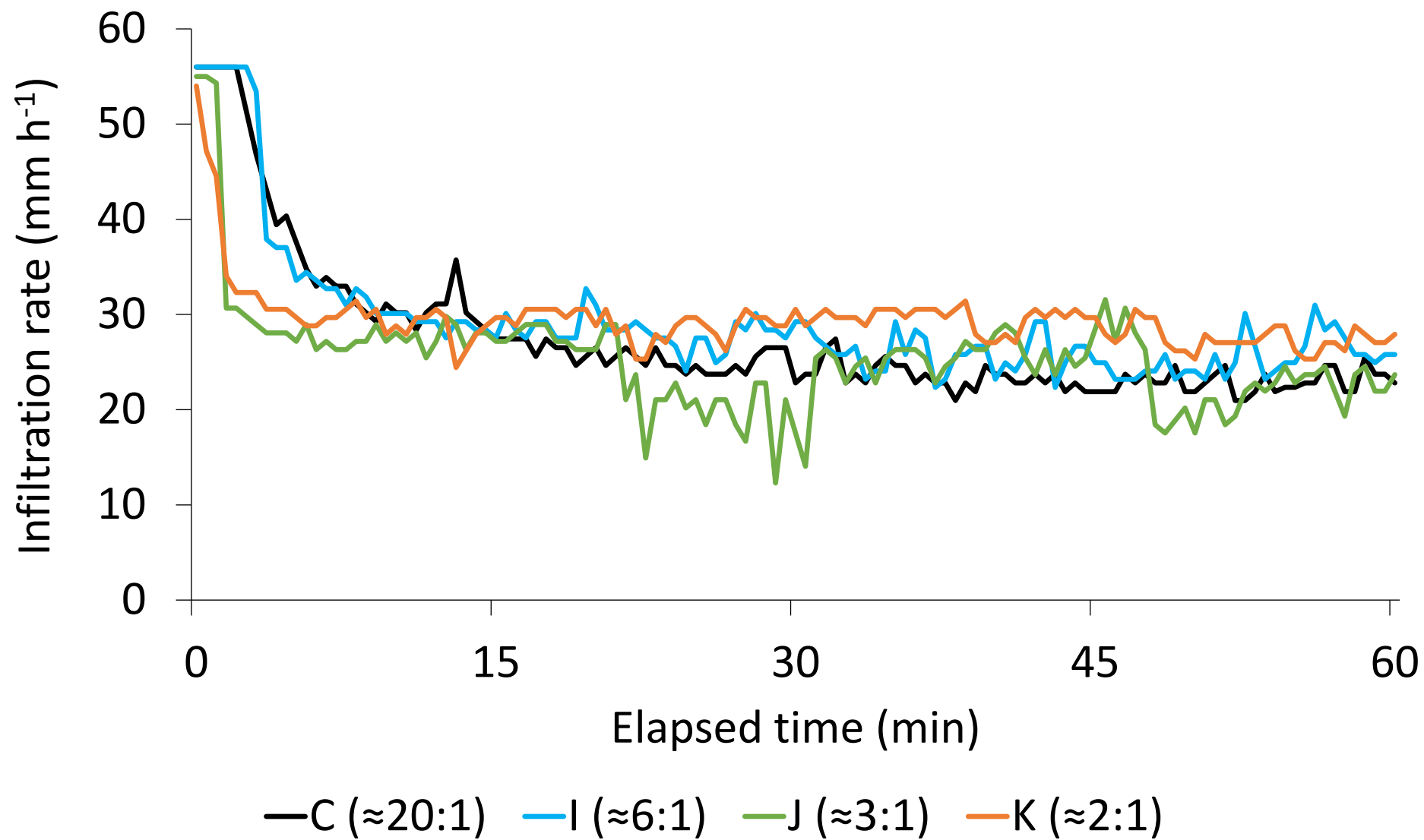
1. Compaction
2. Soil moisture
3. Physical crust
4. Gravel mulch
5. Plant canopy
6. Soil texture
7. Slope











1. Compaction
2. Soil moisture
3. Physical crust
4. Gravel mulch
5. Plant canopy
6. Soil texture
7. Slope

About the same!



How did you do?

Condition			Correct?
1	Uncompacted	Compacted	✓
2	Dry	Wet	✓
3	Crust	No crust	✓
4	Bare soil	Gravel mulch	✓
5	Plant canopy	Bare soil	✓
6	Fine	Coarse	✓
7	Steep	Less steep	✓
Number of correct answers:			7

Modeling infiltration

Physically-based model

- Start with physical laws
- Apply simplifying assumptions to solve Richard's equation

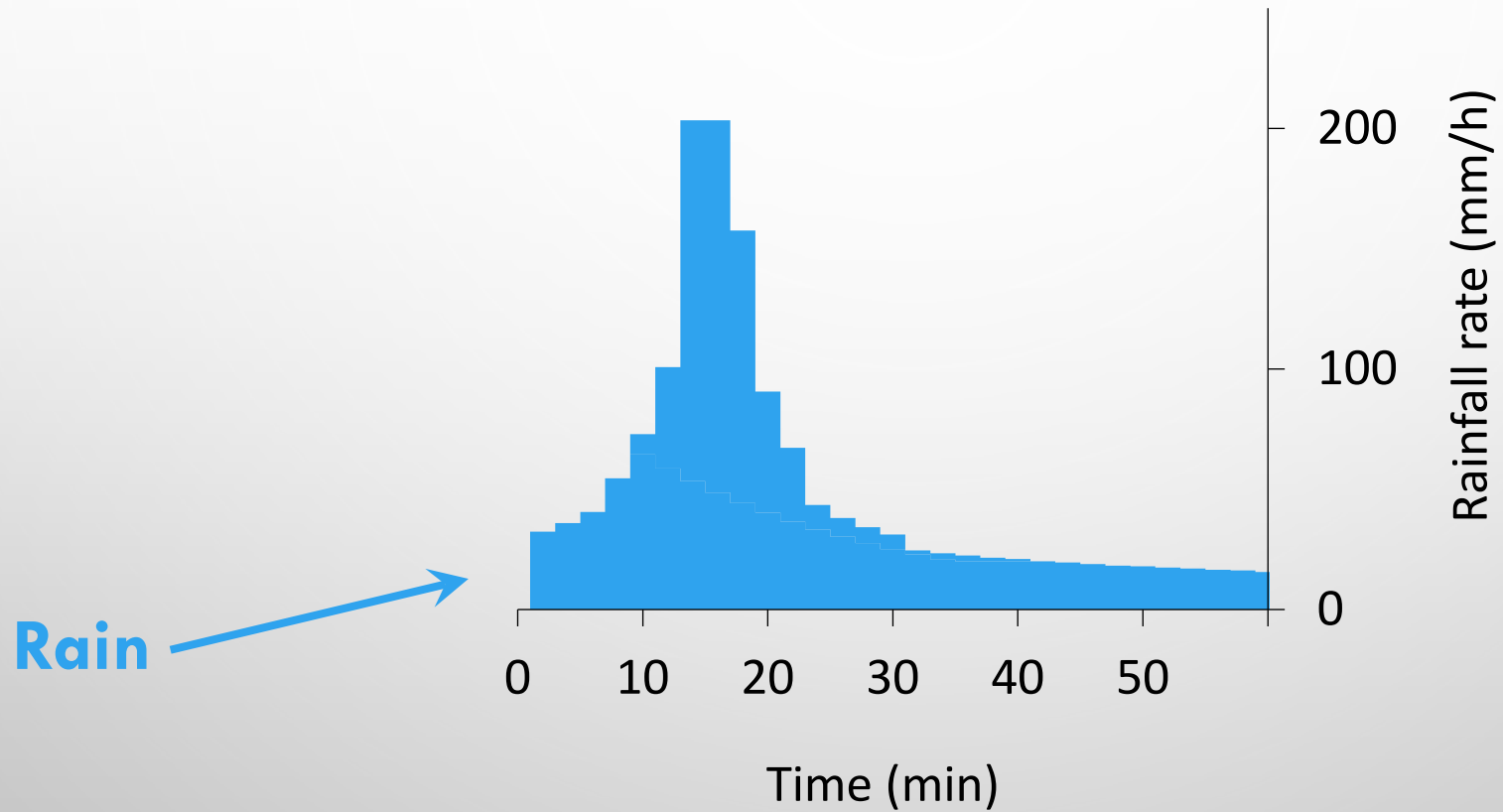
Conceptual model

- Start with conceptualization of core processes
- Find equations to describe processes

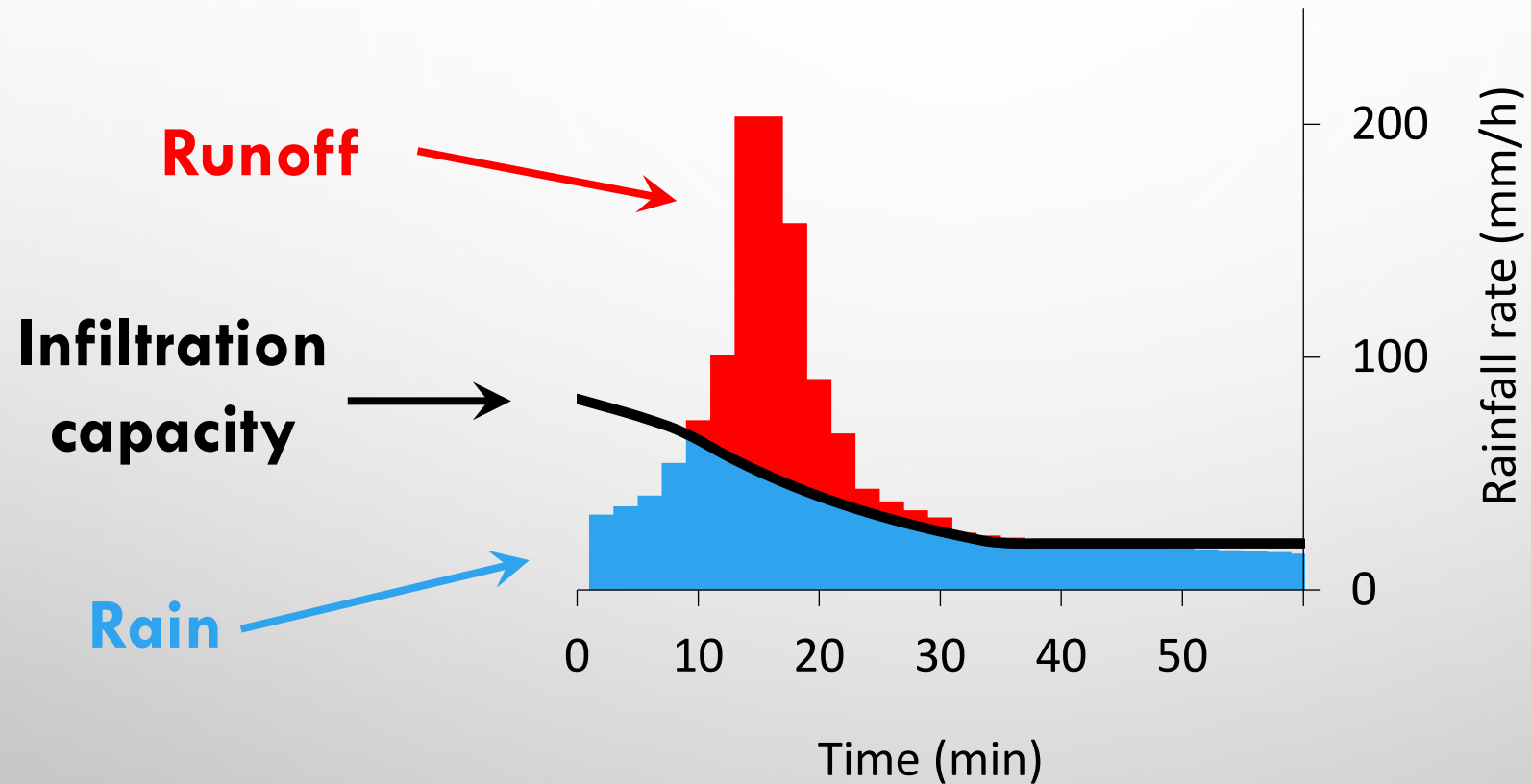
Empirical model

- Based on observations

Modeling infiltration



Modeling infiltration



Modeling infiltration

Physically-based
model

Conceptual
model

Empirical
model

Curve
Number

Initial &
Constant

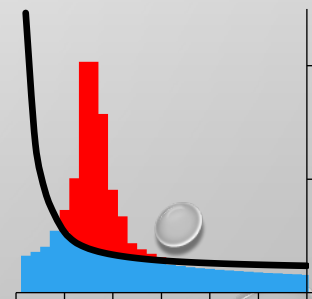
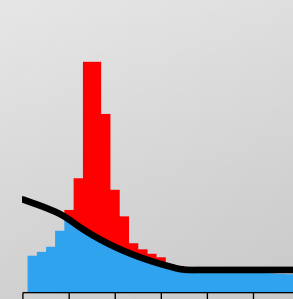
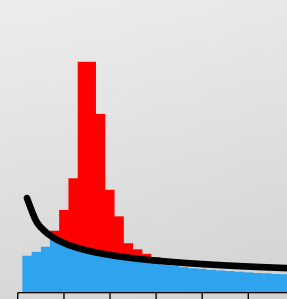
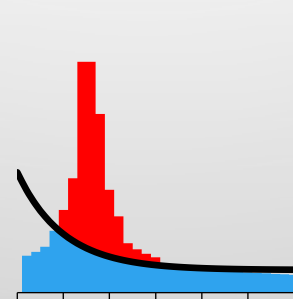
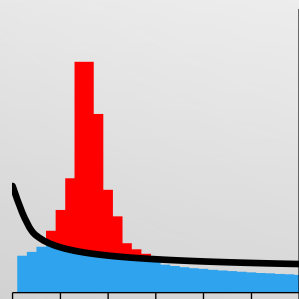
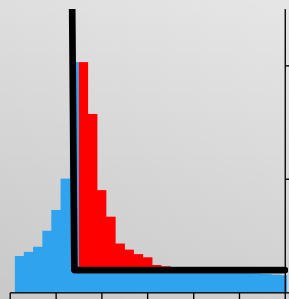
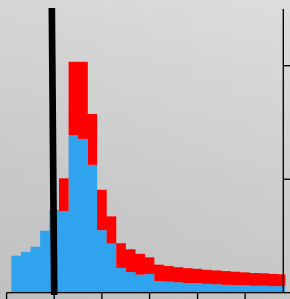
Philip

Horton

Kostiakov

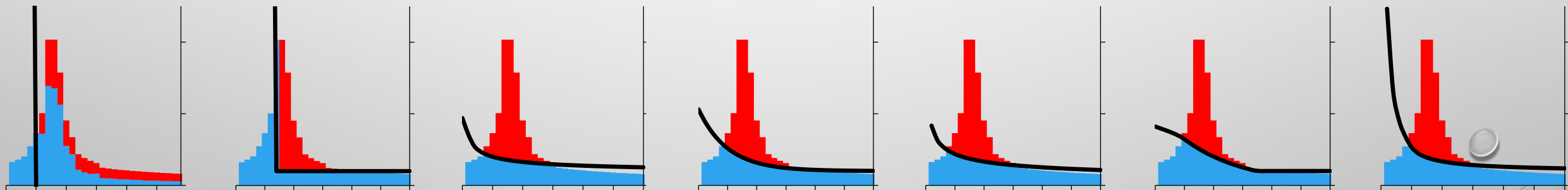
Linear &
Constant

Smith-
Parlange

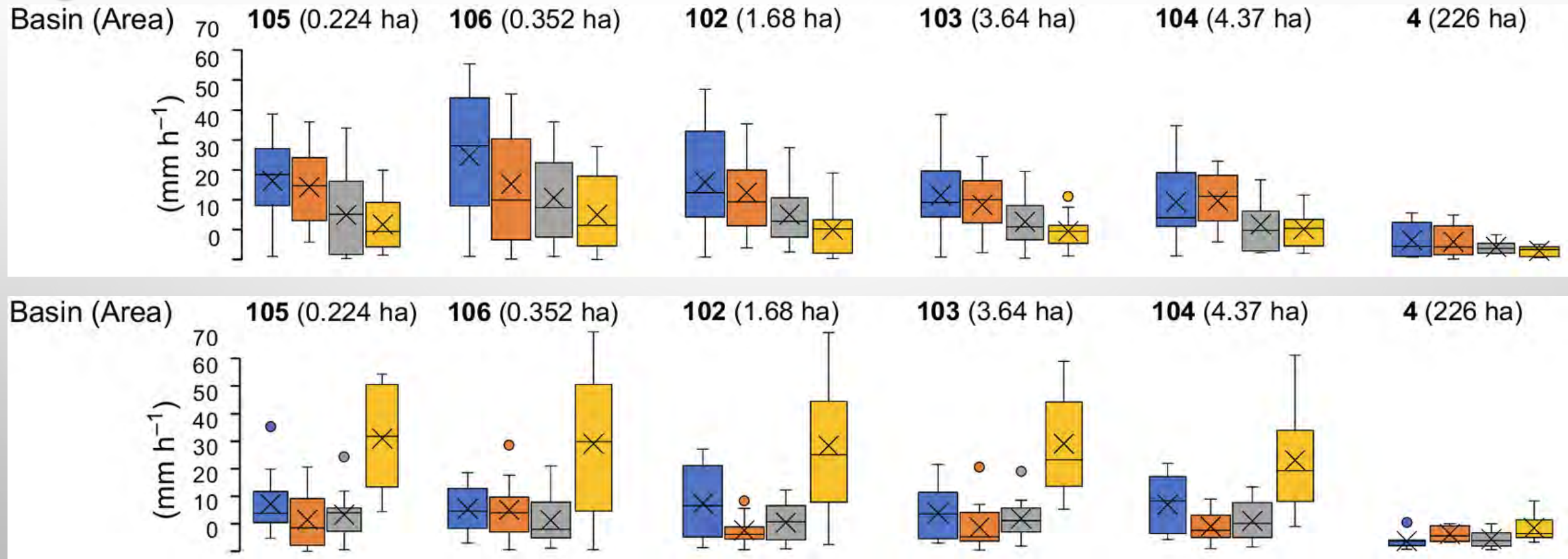


Schoener, G., Stone, M. C., & Thomas, C. (2021). Comparison of seven simple loss models for runoff prediction at the plot, hillslope and catchment scale in the semiarid southwestern US. *Journal of Hydrology*, 598, 126490.

Which one to choose???



Which one to choose???



Which one to choose???

- Accuracy
- Parsimony
- Data availability
- Other constraints

The background image is a wide-angle photograph of a braided river system in a desert environment. The river consists of multiple channels of light-colored, silty water that meander and split across a sandy, sparsely vegetated landscape. In the distance, a range of rugged mountains is visible under a clear blue sky with a few wispy clouds. The overall scene suggests a focus on hydrology or environmental engineering.

EPS 576 – Fall 2025

Questions?

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